

Final Report

Comparative Evaluation of Machine Learning Classifiers for Gallbladder Cancer Detection in Ultrasound images

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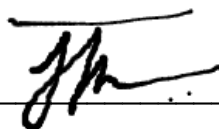
COMP3931 Individual Project

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<i>Final Report</i>	<i>PDF file</i>	<i>Uploaded to Minerva (29/04/26)</i>
<i>Link to online code repository</i>	<i>URL</i> https://github.com/JosephAbraham01/Gallbladder-Ultrasound-Machine-Learning	<i>Sent to supervisor and assessor (29/04/26)</i>
<i>README file</i>	<i>In the Github Repository</i>	<i>Sent to supervisor and assessor (29/04/26)</i>
<i>OneDrive link to GBCU dataset files, only accessible by UoL organisation accounts</i>	<i>Link to .zip file</i> GBCU.zip	<i>Sent to supervisor and assessor (29/04/26)</i>

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Summary

This project investigates wheather machine-learning techniques could be used to improve the classification of gallbladder ultrasound images into normal, benign and malignant classes. This is important because gallbladder cancer is difficult to detect early, due to malignant and benign cases presenting very similarly on ultrasound imaging. Another important aspect is that ultrasound images are often noisy and low quality, which may make image based classification more challenging.

To investigate this a full classification pipeline was developed using the GBCU dataset. Three feature extraction methods (GLCM, LBP, PCA) were implemented and then combined with four classifiers (K-NN, SVM, Decision Tree, Random Forest). Alongside these classical methods a more modern, ResNet-18 CNN was also implemented and trained using different epoch counts.

The models were evaluated using accuracy, precision, recall, F1-score, confusion matrices and ROC curves. The results showed that PCA was the strongest feature extraction method overall, producing better results than GLCM and LBP. The best classical result was achieved by PCA-SVM which gave the highest accuracy. The best deep learning results were achieved by ResNet-18 trained for 50 epochs and produced the highest F1-Score as well as the most balanced performance across metrics.

Every model showed difficulty separating benign and malignant cases, due to their presentation being so similar. Overall, the project met its aim of comparing classical and deep-learning methods and showed that machine learning can successfully identify useful patterns in ultrasound images, evidenced by the strong separation of normal and benign classes. However, in the context of detecting gallbladder cancer, a much larger dataset and more advanced models would be needed to reliably differentiate between benign and malignant cases.

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Chapter 1

Introduction and Background Research

1.1 Introduction

Gallbladder cancer is a rare but highly aggressive disease, often associated with poor treatment due to its late-stage diagnosis. Only 20% of cases are diagnosed in the early stages. Early diagnosis is critical for survival rates, as the five-year survival rate for metastasised gallbladder cancer is 2% compared to 66% for early diagnoses (Cleveland Clinic, 2023). Ultrasound imaging is widely used as a first-line diagnostic tool for gallbladder assessment due to it being non-invasive and more cost-effective than blood tests and MRI scans (Ralls et al., 1985). Despite this, ultrasound images are inherently difficult to interpret due to the images being noisy, low contrast and low quality (Acharya et al., 2012).

Machine learning techniques have become increasingly relevant in medical image analysis, promising the ability to automatically detect diseases through pattern recognition.

Classification algorithms have been successfully used to differentiate between benign and malignant tumours (Litjens et al., 2017). However, this was achieved using CT, MRI scans and standard photography, which typically have much higher image quality compared to ultrasound. As a result, the application of machine learning to ultrasound data presents more challenges than typical cases.

This project aims to investigate and evaluate the suitability of both classical and modern machine learning approaches for the classification of gallbladder ultrasound images. The models will be evaluated using metrics such as classification accuracy, precision, recall and F1 score.

1.2 GallBladder Cancer and Ultrasound Imaging

1.2.1 GallBladder Cancer

According to (Hundal and Shaffer, 2014), gallbladder cancer is a malignant tumour that originates in the epithelial lining of the gallbladder. The disease is relatively rare, accounting for around 1% of all cancers globally. Despite its low occurrence, it has a particularly poor prognosis, with less than 2% of people living beyond 5 years after diagnosis (Cleveland Clinic, 2023).

A major challenge in managing gallbladder cancer is the difficulty of early detection. In its early stages, the disease often presents with subtle morphological changes such as slight wall thickening, small tumours, or minor irregularities in tissue structure. (Sarangi et al., 2025) reported that these features are not unique to malignant cases and frequently resemble benign abnormalities such as gallstones. Therefore, distinguishing between benign and malignant cases is inherently difficult based on visual inspection alone.

Studies such as (Graff, 2016) have shown that early-stage gallbladder cancer is difficult to detect even with modern imaging techniques, often due to the non-specific nature of its presentation. Furthermore, many cases are only identified accidentally during surgical procedures rather than through preoperative imaging, highlighting the limitations of current diagnostic approaches. The lack of clear visual biomarkers suggests that diagnosis relies on recognising subtle combinations of features rather than a single clear indicator.

This is an important motivation for the project. If the relevant visual cues are subtle, overlapping, and difficult to interpret consistently, then the task is well-suited to computational pattern recognition. Machine-learning methods can analyse combinations of texture, intensity, and structural features that may be too fine-grained or too complex to find reliably by eye alone.

1.2.2 Ultrasound Imaging for Gallbladder Diagnosis

Ultrasound imaging uses high-frequency sound waves to create live images of organs, and (Pinto et al., 2013) describe it as the most commonly used first-line method for evaluating gallbladder abnormalities due to its non-invasive nature, low cost, and widespread availability. It allows clinicians to visualise features such as gallstones, wall thickening, and masses in real time without exposing patients to ionising radiation. These characteristics make it more suitable than CT or MRI for routine initial assessment and in lower-resource settings.

Despite these advantages, ultrasound imaging presents several significant challenges. (Singh et al., 2023) state that the quality of ultrasound images is inherently limited by speckle noise, an interference pattern caused by the collision of sound waves, which reduces image clarity and obscures fine details. In addition, ultrasound images typically have lower contrast and resolution compared to CT and MRI scans, making subtle abnormalities more difficult to detect. (Zhao et al., 2019) further argue that diagnostic accuracy in ultrasound depends heavily on radiologist experience and is affected by speckle noise. In practical terms, this means that diagnostically relevant details may be obscured or interpreted inconsistently across clinicians.

This matters because ultrasound struggles to reliably distinguish between benign and malignant gallbladder abnormalities. Research has shown that features such as gallbladder wall thickening, which may indicate early-stage cancer, are visually similar to benign conditions and cannot be consistently differentiated using ultrasound alone (Sarangi and Kumar, 2025). Additionally, false positive rates in the detection of gallbladder tumours can be high, further complicating diagnosis. (Sarangi and Kumar, 2025) found that the sensitivity of ultrasound for detecting gallbladder cancer is as low as 58.3%, particularly in the early-stage, where features are less pronounced. This indicates that a significant proportion of positive cases may be missed during initial screening.

Other imaging techniques, such as Computed Tomography (CT) and Magnetic Resonance Imaging (MRI), are also used in the diagnosis of gallbladder cancer. (Furlan et al., 2008) reported that CT scans provide higher-resolution cross-sectional images and are more effective for assessing tumours, while MRI offers superior soft-tissue contrast, enabling better detection of subtle gallbladder wall thickening. However, both CT and MRI are more expensive, less accessible, and, in the case of CT, involve exposure to ionising radiation (Barakos et al., 1999).

Taken together, the literature suggests that ultrasound is the correct modality to target for improvement as it is already the standard first-line test, but its diagnostic value is constrained by noise, low contrast, and operator dependence. This creates a strong case for applying machine learning to improve classification accuracy.

1.2.3 Why Ultrasound Still Matters

Despite those weaknesses, ultrasound remains the most practical method to improve through machine learning. If CT and MRI are more informative but less available, then enhancing ultrasound has potentially greater real-world value (Rana et al., 2024). A successful ML system could make a common, low-cost test more reliable, rather than depending on more expensive imaging for every suspicious case. From a project perspective, this makes ultrasound a strong application domain as it is clinically relevant, technically challenging, and well aligned with automated image classification.

1.3 Machine Learning for Medical Image Analysis

Machine Learning for medical image analysis uses classification algorithms that use underlying patterns and the relationship between features in the data to classify new data. In supervised learning, models are trained using labelled data to learn the relationship between features and the output classes (Litjens et al., 2017). Haenssle et al. (2018) investigated its effectiveness in detecting melanoma and found that modern Convolutional Neural Networks

(CNN's) outperformed the average dermatologist. However, it's important to take into account that the images used in the melanoma study were of higher quality with less noise than ultrasound, making it easier to obtain a good model. Basu et al. (2022), in the CVPR paper introducing GBCNet for gallbladder cancer detection from ultrasound, explicitly note that ultrasound images suffer from low image quality, significant noise, and varying viewpoints caused by the handheld sensor. Their work is important because it is gallbladder-specific and demonstrates that image-classification methods struggle unless the region of interest is preprocessed. Techniques such as Grey-Level Co-occurrence Matrix (GLCM) and Local Binary Patterns (LBP) enable the extraction of texture features that describe spatial relationships between pixels, enabling more detailed analysis of tissue characteristics. This helps make the diagnostic process less subjective and more consistent.

1.3.1 Best Machine Learning Approaches in Medical Imaging

Litjens et al. (2017) and more recent reviews of clinical AI both suggest that deep learning, particularly CNN-based architectures, has become the dominant approach in medical image analysis because it can learn hierarchical features directly from image pixels. This is one reason why a CNN-based model should be included in the project. At the same time, they also note that traditional approaches remain useful where data is limited, like in this project's case. The literature, therefore, does not justify assuming that CNNs will always outperform classical methods on ultrasound; instead, it justifies comparing both. Zhao et al. (2019) implies that the "best" approach for ultrasound is whichever model can best handle noisy and texture-dominated data. This provides a strong rationale for comparing a modern CNN with texture-based classical pipelines.

1.4 Machine Learning Approaches

1.4.1 Support Vector Machine(SVM)

Support Vector Machines (SVMs) classify data by finding an optimal hyperplane that separates different classes. This is a boundary that neatly separates data points. Through the use of kernel functions, SVMs can model non-linear relationships in data, making them suitable for classification tasks such as medical image analysis (Jiang et al., 2020).

1.4.2 k-Nearest Neighbours

k-Nearest Neighbours (k-NN) classifies new data based on the average class of their nearest neighbours in the feature space. The similarity is measured using the Euclidean distance between points and the unlabelled data. Its performance depends heavily on how well extracted features separate between normal, benign, and malignant cases. (Zhang, 2016).

1.4.3 Decision Trees

Decision trees classify data through a series of hierarchical nodes, each of which is a decision/question based on the features that were used to train the model. Each branch represents an outcome of a decision; the root node is the starting point for new data that needs to be classified, and the leaf nodes represent the final predicted class for the data (Kingsford and Salzberg, 2008).

1.4.4 Random Forest

Random Forest is an ensemble method that combines multiple Decision Trees to improve generalisation performance. Each tree has different nodes and therefore can have different predictions. The most popular prediction among the trees is selected. By averaging predictions across multiple trees, it reduces overfitting and is robust to noise, making it particularly suitable for ultrasound data (Breiman, 2001)

1.4.5 Deep Learning Approach

The other classifiers are considered classical approaches, so to properly evaluate the use of machine learning on ultrasound data, I will also use a more modern approach to image classification.

Convolutional Neural Networks (CNNs) are widely used for image classification tasks due to their ability to automatically learn hierarchical features from raw image data. Unlike classical approaches, CNNs do not require manual feature extraction; instead learning relevant patterns directly from pixel intensities (Krizhevsky et al., 2012).

ResNet-18 is a CNN architecture that uses residual connections to enable effective training of deeper networks by mitigating the vanishing gradient problem (He et al., 2016). Neural networks update parameter weights using back propagation, which causes the data to diminish at each layer (vanishing gradient). Residual connections skip layers to avoid this. This makes it a suitable modern approach for comparison with classical machine learning methods in this project. A critical point, however, is that deep learning requires more data and computation than the classical models above

1.5 Feature Extraction Techniques

Classical machine learning approaches use feature extraction techniques to isolate the most important information from the raw data (Jolliffe, 2002). This is an extremely important aspect of the project, as different combinations of feature extraction methods and machine learning approaches can produce very different prediction accuracies for the same machine

learning approach. It's important to take these into account when evaluating the use of machine learning for classifying ultrasound images.

This is especially useful for ultrasound images, as (Noble and Boukerroui, 2006) found that ultrasound images suffer from speckle noise and low contrast, making pixel-based analysis challenging, so feature extraction avoids feeding irrelevant noise to the classifier by first extracting meaningful data, such as texture features and shapes, to highlight the difference between normal tissue and tumours. This allows the classifiers to better find relevant patterns.

1.5.1 Grey-Level Co-occurrence Matrix (GLCM)

GLCM captures texture information by analysing the spatial relationships between pixel intensities. (Haralick et al., 1973) found that it was effective for distinguishing different tissue patterns, making it a good choice for differentiating normal tissue from benign and cancerous growths.

1.5.2 Local Binary Patterns (LBP)

LBP describes textures in an image by taking a pixel's neighbours and comparing them to the centre pixel to get a new value. When you do this across the whole image, you get a pattern that describes the texture of the image. This would be useful for differentiating the texture of a tumour and regular tissue. (Ojala et al., 2002).

1.5.3 Principal Component Analysis (PCA)

PCA reduces dimensionality by transforming features into principal components that capture the most variance in the data. It reduces the complexity of the data while keeping the most important information. Ultrasound images have a lot of data, where each pixel is a feature. Most of these features are not important, and PCA can be used to reduce the data down to its meaningful patterns and remove noise (Jolliffe, 2002).

1.6 Existing Solutions and Research Gap

Basu et al. (2022) demonstrate that deep learning can achieve strong results on gallbladder ultrasound images, especially when ROI selection is integrated into the pipeline. However, the literature is less clear on how classical feature-based pipelines compare directly with modern CNNs on the same task.

This creates a meaningful research gap. There is value in understanding which combination of preprocessing, feature extraction, and classifier design is most appropriate for this application. This is especially relevant for smaller-scale or resource-constrained settings,

where classical ML may remain attractive if it offers reasonable performance, lower computational cost, and greater interpretability.

1.7 Risk Analysis

The project's limited dataset size can reduce generalisability and increase overfitting risk, especially for deep models. This is relevant here because CNN performance is sensitive to data volume and variation.

(Sarangi et al.,2025) show that benign and malignant gallbladder abnormalities may be visually similar on ultrasound. This creates a class-overlap risk, meaning that false negatives and false positives must both be taken seriously in evaluation.

1.8 Project aim, Objectives and Deliverables

1.8.1 Project Aim

This project aims to investigate and compare the effectiveness of classical machine learning and deep learning techniques for the classification of gallbladder ultrasound images into normal, benign, and malignant categories.

1.8.2 Objectives

- To preprocess ultrasound images and use region-of-interest annotations to focus analysis on the gallbladder region
- To implement GLCM, LBP, and PCA as feature extraction strategies.
- To train and evaluate k-NN, SVM, Decision Tree, and Random Forest classifiers in combination with the feature extraction techniques.
- To implement a ResNet-18 CNN as a modern comparison model.
- To evaluate models using accuracy, precision, recall, F1-score, confusion matrices, and ROC curves.

1.8.3 Deliverables

- The results of the machine learning training and testing. Including tables for each model showing their Accuracy, F1-Score, Precision and Recall. A confusion matrix for visualisation of their performance, as well as ROC curves.
- A GitHub repository which will contain the code used to make and evaluate each model

Chapter 2

Methodology

2.1 Designing the Solution

The solution was structured as an end-to-end pipeline consisting of dataset loading, region-of-interest extraction, preprocessing, feature extraction or direct image input, model training, and evaluation. This design reflects the standard structure of medical image classification studies, where preprocessing, representation learning, and classification are treated as distinct but linked stages (Litjens et al., 2017). Having a standardised pipeline would also allow for a fair comparison of models.

A particularly important design choice was to use the gallbladder region of interest (ROI) rather than the full ultrasound frame. (Basu et al., 2022) demonstrated that strong results can be achieved on gallbladder images when ROI selection is part of the pipeline in deep learning. The official description of the GBCU dataset, which I will be using, states that expert radiologists drew axis-aligned bounding boxes covering the gallbladder and adjacent liver parenchyma for each image, providing a clinically meaningful localisation of the relevant region (IIT Delhi, 2022). Since ultrasound images often contain irrelevant noise and other body parts, using bounding boxes helps prevent the classifier from learning unwanted correlations. The datasets' readme also confirms that "bbox_annot.json" contains the radiologist-drawn bounding boxes and that "train.txt" and "test.txt" provide the image-level labels used for classification.

2.1.1 Machine Learning Models

When selecting the machine-learning models for the project, three criteria were used.

- The models needed to support multiclass classification, because the dataset labels were normal, benign, and malignant.
- They needed to be suitable for moderate-sized medical datasets (1255 images), since the project was not based on a very large image collection.
- They needed to represent a range of modelling strategies so that the comparison would be meaningful rather than redundant.
- Must have proven good performance in medical image analysis

The models needed to support multiclass classification. This requirement arose directly from the structure of the dataset, which classifies each ultrasound image as normal, benign, or

malignant rather than as a simple binary cancer/non-cancer outcome. As a result, models were required to distinguish between clinically similar but distinct categories, particularly benign and malignant abnormalities, which is a more demanding task than binary classification. Selecting models that could naturally handle multiclass prediction ensured that the experimental design matched the real structure of the problem rather than oversimplifying it.

The models needed to be suitable for moderate-sized medical datasets. Although the GBCU dataset is substantial, it is still relatively small compared with the very large datasets often used to train high-capacity deep-learning systems. The selected classical models needed to be capable of learning effectively from limited to moderate amounts of labelled data without depending on extremely large sample sizes. This criterion helped ensure that the methods chosen were realistic for the scale of the project.

The models needed to represent a range of modelling strategies so that the comparison would be meaningful rather than redundant. If several very similar models had been selected, the evaluation would provide limited insight into how different classification principles behave on gallbladder ultrasound images. By choosing models based on different underlying ideas, such as distance-based classification, tree-based classification, and ensemble learning, the project could compare not only performance but also the suitability of different modelling paradigms for noisy medical image data.

Based on these criteria, four classical classifiers were selected:

- k-Nearest Neighbours (k-NN) (Zhang, 2016)
- Support Vector Machine (SVM) (Jiang *et al.*, 2020)
- Decision Tree (Kingsford and Salzberg, 2008)
- Random Forest (Breiman, 2001)

A ResNet-18 Convolutional Neural Network was also implemented. CNNs were included because the medical imaging literature identifies deep learning as a leading approach for image classification when data and preprocessing are appropriate (Litjens *et al.*, 2017). ResNet-18 was chosen specifically because it is a relatively lightweight network that still provides enough power for image classification while being feasible on limited hardware and data (He *et al.*, 2016).

2.1.2 Feature Extraction Techniques

Three feature extraction techniques were selected for the classical machine-learning pipeline: GLCM, LBP, and PCA.

GLCM was chosen because ultrasound classification relies heavily on texture, and it is a well-established statistical method for representing spatial relationships among pixel intensities. This made it suitable for distinguishing among normal tissue, benign abnormalities, and malignant growths (Haralick *et al.*, 1973).

LBP was selected because it captures local texture patterns efficiently and therefore provides a useful contrast to GLCM. While GLCM describes broader statistical texture relationships, LBP focuses on local micro-texture, making it valuable for comparison on noisy ultrasound ROI images (Ojala *et al.*, 2002).

PCA was included because it provides a different type of representation from the two texture-based methods. Rather than explicitly describing texture, PCA reduces dimensionality by compressing image information into a smaller set of components. This made it useful for testing whether a reduced representation of the ROI could still support accurate classification while lowering redundancy and complexity (Jolliffe, 2002).

2.1.3 Chosen Frameworks and Tools

Python was chosen as the main development language because it provides libraries for machine learning (both classical and deep learning), image processing, and model evaluation. It also allowed all stages of the project, including preprocessing, model training and evaluation to be implemented in the same environment.

For image processing, OpenCV was used to load images, convert them to grayscale where appropriate, crop regions of interest, resize images, and perform basic preprocessing operations. OpenCV was chosen because it provides efficient, reliable image-processing functions and integrates well with NumPy arrays for the input feature matrix and the list of labels.

scikit-image was used for GLCM and LBP feature extraction, as it provides standard implementations of these methods and reduces the risk of errors from custom implementations for each model. This makes the experiments more reproducible.

scikit-learn was used for PCA, the four classical classifiers, and the evaluation metrics. This was appropriate because it provides a consistent and well-documented framework for classical machine learning, allowing the same workflow to be applied across models. It also ensured that the values for metrics such as accuracy and precision were calculated correctly, making the results more reliable.

PyTorch and torchvision were used for the ResNet-18 implementation. PyTorch was chosen because it supports transfer learning and provides enough flexibility to define a custom

ultrasound dataset based on ROI-cropped images. Torchvision was used to load the pretrained ResNet-18 architecture.

Jupyter notebooks were used during the experimentation stage to build the models and evaluate them using metrics, plots and confusion matrices because the inherent separation of the code into cells made it easier to visualise results and compare models. Separating the different stages of the pipeline into different cells made the code easier to visualise and understand, as well as debug.

2.1.4 Resulting System Design

The resulting design consisted of two analytical pipelines.

Classical machine-learning pipeline: In this pipeline, each ultrasound image was loaded and processed using OpenCV, and the gallbladder ROI was extracted using the radiologist-provided bounding box annotations from “bbox_annot.json”. The cropped ROI was then resized to a fixed size using OpenCV. After this, a feature extraction method was applied using either scikit-image for GLCM and LBP, or scikit-learn for PCA. The resulting feature vector was then passed into one of the four classical classifiers, implemented in scikit-learn: k-NN, SVM, Decision Tree, or Random Forest.

deep-learning pipeline: In this version, the ROI image was again loaded and cropped using OpenCV, then resized and normalised before being passed directly into ResNet-18, implemented using PyTorch and torchvision. This allowed the model to learn feature representations directly from the image data rather than relying on handcrafted features.

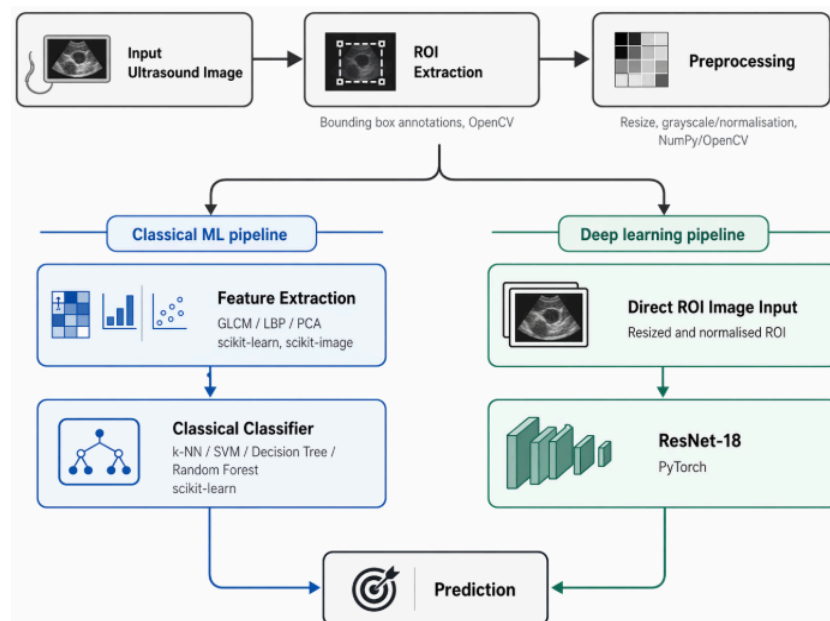


Figure 1: Classical and deep learning pipeline diagram

2.2 Development

2.2.1 Development Management Approach

An Agile development approach was followed for this project (Beck *et al.*, 2001). A short sprint-based structure was used to divide the project into manageable stages with corresponding deliverables. Agile was chosen because the project involved uncertainty in several areas, including feature extraction effectiveness, model performance, and CNN training behaviour. A rigid linear plan would have made it difficult to react to experimental findings, whereas an iterative approach allowed me to reflect on my findings at the end of a sprint and plan the next one based on my evaluation. (Dybå and Dingsøy, 2008).

Development was organised into three 2-week sprints. Short sprint timelines were used because they encourage clear, achievable goals while maximising productivity (Highsmith, 2009). Agile also emphasises the delivery of working components early in development, which suited the project well, as it allowed a baseline preprocessing and classification pipeline to be built before other feature extraction experiments were done (Beck *et al.*, 2001).

Project tasks were organised using a kanban board with the categories To Do, In Progress, Backlog and Done. This allowed transparent visibility of the progress of each sprint (VersionOne, 2020).

The main project milestones were:

- successful loading and preparation of the GBCU dataset, including ROI extraction
- Creation of a baseline pipeline using one of the Feature Extraction techniques
- completion of the classical machine-learning pipeline using GLCM, LBP, and PCA
- full comparative evaluation of k-NN, SVM, Decision Tree, and Random Forest
- successful implementation of the ResNet-18 CNN and evaluation across increasing epoch counts
- Completion of the final comparative analysis of all approaches

The project was divided into the following sprints:

- **Sprint 1 (2 weeks):** establish the baseline pipeline, including loading the GBCU dataset, extracting the ROI from the provided bounding boxes, preprocessing the images, and training the initial classical machine-learning models.

- **Sprint 2 (2 weeks):** carry out controlled experimentation with the three feature extraction techniques and the four classical classifiers, then compare their performance using the chosen evaluation metrics.
- **Sprint 3 (2 weeks):** implement and evaluate the ResNet-18 CNN using increasing numbers of epochs, then complete the final comparative analysis between the classical and deep-learning approaches.

2.2.2 Version Control and Good Practice

The code developed for this project was version-controlled using Github, which provided a clear record of changes and made it easier to manage different stages of development. This was particularly useful because the project involved several alternative modelling pipelines, based on different feature extraction techniques used in the classical machine-learning experiments. The use of Github helped keep experimentation organised and reduced the risk of losing working versions of the code (Chacon and Straub, 2014).

Separate branches were created for the main feature extraction approaches, allowing each technique to be developed and tested independently before being merged into the main branch. This helped keep the experimental pipelines separate while they were being refined. Descriptive commits were used to easily track changes and revert to previous versions if necessary. Link to Github repository on **(page ii)**.

Python code was written in line with PEP 8 style guidelines, using consistent formatting, indentation, and naming conventions. Comments were added where needed to explain important preprocessing, feature extraction, and modelling steps. Simple error checks, such as handling missing files or invalid ROI crops, were also included to make the code more reliable during repeated experiments.

2.2.3 Data Collection

The project used the Gallbladder Cancer Ultrasound (GBCU) dataset, which the official dataset page describes as the first public dataset for gallbladder cancer identification from ultrasound images (IIT Delhi, 2022). According to the dataset authors, GBCU contains 1,255 annotated abdominal ultrasound images from 218 patients, with class counts of 432 normal, 558 benign, and 265 malignant. The dataset authors also state that the training set contains 1,133 images and the testing set 122 images, and that all images from a given patient are placed entirely in either the training or testing split to preserve patient-level separation (IIT Delhi, 2022).

The same source states that the images were collected from patients referred to PGIMER, Chandigarh hospital, for abdominal ultrasound examinations of suspected gallbladder

pathologies, and that the data were fully anonymised following ethics approval and informed consent procedures (IIT Delhi, 2022). This means the dataset is clinically relevant and directly aligned with the task of finding gallbladder abnormalities.

A major advantage of GBCU is that it provides both class labels and expert-defined ROI annotations. The official dataset description states that each image is labelled as normal, benign, or malignant and that expert radiologists drew an axis-aligned bounding box. This validates that the labels and ROI's are reliable.

2.2.4 Data Preparation

Image names were matched to their class labels using the provided training and testing files. The gallbladder ROI was then extracted from each image using the bounding box annotations and resized to a fixed size for modelling. Grayscale ROIs were used for classical feature extraction, while the ResNet-18 pipeline used resized and normalised ROIs to match the pretrained CNN input requirements. Focusing on the ROI reduced the risk of the models learning irrelevant background information.

2.2.5 CNN Training Strategy

Rather than using a single arbitrary epoch count, the ResNet-18 model was trained with increasing numbers of epochs, and the results were compared. This was done because training for too few epochs may limit learning, while excessive training can lead to overfitting, especially on limited image datasets (Goodfellow, Bengio and Courville, 2016). Evaluating the model at different training durations, therefore, made it possible to assess whether additional epochs improved generalisation or only reduced training loss.

This made the CNN experiment more informative than training once and reporting a single result, since it allowed the effect of training duration to be analysed directly.

2.2.6 Evaluation Design

Performance was measured using accuracy, precision, recall, F1-score, confusion matrices, and ROC curves because accuracy alone would not be sufficient in a medical classification problem. Precision and recall provide insight into false positives and false negatives, which would be important if the models were ever used in a clinical setting. F1-score gives a balanced summary of both, confusion matrices visually show how errors are distributed across the three classes, and ROC curves show how well the models distinguish between classes. (Akben, 2025) similarly uses this combination of metrics to give a more complete evaluation in the context of using classical machine learning approaches to predict gallstones in ultrasound images.

Chapter 3

Implementation and Validation

3.1 Sprint 1

3.1.1 Implementation

The aim of Sprint 1 was to establish a working baseline pipeline using the GBCU dataset, ROI extraction, and GLCM-based feature extraction. The reason GLCM was used as the baseline is that studies such as (Haralick *et al.*, 1973) show that texture is the most diagnostically relevant feature in ultrasound imaging, making GLCM an appropriate starting point for a successful pipeline. This stage was important because it created the complete workflow that would be used to help standardise later experiments, making them more reliable.

The first implementation step was to load the radiologist-assigned image labels from the provided “train.txt” and test.txt files and match them to the corresponding ultrasound images.

```
def load_labels(path):
    labels = {}
    with open(path) as f:
        for line in f:
            img, lbl = line.strip().split(',')
            labels[img] = int(lbl)
    return labels
```

Figure 2: load in image labels from text files

This was implemented using standard Python file handling because the label files were already in a simple comma-separated format and did not require a more complex data-processing framework such as Pandas at this stage. A dictionary-based representation was chosen because it provided efficient lookup between image names and class labels.

The ROI annotations were then loaded from “bbox_annot.json”. The JSON format made this straightforward to implement. Using the radiologist-defined ROI was a deliberate design choice, since the project aimed to classify patterns in the gallbladder itself rather than in the surrounding abdomen. The ROI extraction and resizing were implemented using OpenCV.

```
def extract_features(img_path, bbox):
    img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)

    x1, y1, x2, y2 = bbox
    roi = img[y1:y2, x1:x2]

    roi = cv2.resize(roi, (128, 128))
```

Figure 3: load in Region of interest(ROI) annotations and resize images

OpenCV was chosen because it provides fast and reliable image-cropping and resizing functions, which are central to ultrasound preprocessing. Grayscale loading was used because the GBCU ultrasound images are grayscale, and the GLCM texture representation does not require colour information.

For the baseline feature extraction method, GLCM was implemented using scikit-image.

```
glcm = graycomatrix(  
    roi,  
    distances=[1],  
    angles=[0],  
    levels=256,  
    symmetric=True,  
    normed=True  
)  
  
features = [  
    graycoprops(glcm, 'contrast')[0, 0],  
    graycoprops(glcm, 'homogeneity')[0, 0],  
    graycoprops(glcm, 'energy')[0, 0],  
    graycoprops(glcm, 'correlation')[0, 0]  
)  
  
return features
```

Figure 4: GLCM feature extraction sprint 1 baseline

scikit-image was chosen because it provides a standard and well-tested implementation of GLCM, reducing the likelihood of implementation error.

The extracted features and labels were then assembled into the feature matrix x and target vector y. NumPy arrays were used because they are the standard numerical format expected by scikit-learn

```
x, y = [], []  
  
labels = load_labels("train.txt")  
  
for img_name, lbl in labels.items():  
    if img_name not in bbox_data:  
        continue  
  
    bbox = bbox_data[img_name]["bbs"][0][1]  
    features = extract_features(f"imgs/{img_name}", bbox)  
  
    x.append(features)  
    y.append(lbl)  
  
x = np.array(x)  
y = np.array(y)
```

Figure 5: Creation of feature matrix and target vector

The baseline classifiers were then implemented using scikit-learn.

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, stratify=y, random_state=42
)

models = {
    "kNN": KNeighborsClassifier(n_neighbors=5),
    "SVM": SVC(kernel="rbf", probability=True),
    "Decision Tree": DecisionTreeClassifier(max_depth=5),
    "Random Forest": RandomForestClassifier(n_estimators=100)
}
```

Figure 6: Scikit-learn classifier implementation

scikit-learn was chosen because it provides all of the classical classifiers required by the project, along with functions for training, testing, and evaluation. This made it suitable for comparative experimentation. Unlike deep-learning frameworks such as TensorFlow or PyTorch, scikit-learn is better suited for traditional feature-based models and allows the baseline pipeline to be implemented efficiently.

3.1.2 Validation

The purpose of validation in Sprint 1 was to confirm that the baseline GLCM pipeline was functioning correctly and that the extracted features could support classification. The initial validation focused on checking that the full workflow from ROI extraction to prediction produced sensible outputs, and that the classifiers were learning meaningful distinctions. I used the `classification_report` function from `sklearn.metrics` to output the accuracy, precision, recall and f1-scores for each model

```
for name, model in models.items():
    model.fit(X_train, y_train)
    preds = model.predict(X_test)

    print(f"\n{name}")
    print(classification_report(y_test, preds))
```

Random Forest				
	precision	recall	f1-score	support
0	0.74	0.66	0.70	80
1	0.69	0.79	0.74	102
2	0.70	0.58	0.63	45
accuracy			0.70	227
macro avg	0.71	0.68	0.69	227
weighted avg	0.71	0.70	0.70	227

Figure 7: Classification report usage

Figure 8: GLCM baseline pipeline results

3.2 Sprint 2

3.2.1 Implementation

The aim of sprint 2 was to alter the pipeline by implementing different feature extraction techniques and evaluating them across the four classical classifiers.

The LBP pipeline was implemented using `scikit-image`

```
def extract_lbp_features(img_path, bbox):
    img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)

    if img is None:
        return None

    x1, y1, x2, y2 = bbox
    x1, y1 = max(0, x1), max(0, y1)
    x2, y2 = min(img.shape[1], x2), min(img.shape[0], y2)

    roi = img[y1:y2, x1:x2]
    if roi.size == 0:
        return None

    roi = cv2.resize(roi, (128, 128))

    radius = 1
    n_points = 8 * radius

    lbp = local_binary_pattern(roi, n_points, radius, method='uniform')
    hist, _ = np.histogram(lbp.ravel(), bins=np.arange(0, n_points + 3), range=(0, n_points + 2))

    hist = hist.astype("float")
    hist /= (hist.sum() + 1e-6)

    return hist
```

Figure 9: LBP Feature Extraction implementation

LBP was chosen because it provided a second texture-based representation that contrasted with GLCM. While GLCM captures broader statistical texture relationships, LBP focuses on local neighbourhood patterns. Implementing both allowed texture analysis to be explored from two different perspectives.

PCA was included because it provided a non-texture-based comparison. Unlike GLCM and LBP, which were chosen specifically to describe texture, PCA reduced dimensionality by retaining the main variance in the ROI data. This allowed the project to test whether a compressed representation of image information could still lead to effective classification.

For the PCA-based pipeline, each ultrasound image was first loaded in grayscale, cropped to the gallbladder region using the bounding box annotation, resized to a fixed resolution, and normalised so that pixel values lay between 0 and 1. The ROI was then flattened into a one-dimensional vector, producing a pixel-based representation of each image.

```
def extract_pca_input(img_path, bbox):
    img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)

    if img is None:
        return None

    x1, y1, x2, y2 = bbox
    x1, y1 = max(0, x1), max(0, y1)
    x2, y2 = min(img.shape[1], x2), min(img.shape[0], y2)

    roi = img[y1:y2, x1:x2]
    if roi.size == 0:
        return None

    roi = cv2.resize(roi, (64, 64))
    roi = roi.astype(np.float32) / 255.0

    return roi.flatten()
```

Figure 10: PCA Feature Extraction implementation

Principal Component Analysis (PCA) was then applied to the vectors to reduce their dimensionality, retaining 50 principal components.

3.2.2 Validation

Validation in Sprint 2 focused on comparing the feature extraction techniques fairly across the four classical classifiers. A similar approach is used by (Akben, 2025), who evaluates ultrasound classification models using accuracy, precision, recall, F1-score, confusion matrices, and ROC analysis, supporting the suitability of this metric set for this study

Accuracy was included to provide an overall indication of predictive performance, but it was not treated as sufficient on its own. Precision and recall were particularly important because they show how well the models handled false positives and false negatives. The F1-score was of particular interest because it gives a balanced summary of precision and recall and is useful where both types of error should be minimised. This is important as false negatives and false positives can have serious consequences in a medical setting. Confusion matrices were also generated to show visually how errors were distributed across the classes. ROC curves were used to show how well the model can distinguish between classes. This was important as benign and malignant tumours both presented very similarly.

	Model	Accuracy	Precision	Recall	F1 Score
0	k-NN	0.674009	0.667050	0.641394	0.650692
1	SVM	0.713656	0.747837	0.659286	0.678677
2	Decision Tree	0.665198	0.671212	0.620098	0.631456
3	Random Forest	0.700441	0.706614	0.669826	0.682650

Table 1: GLCM Results Table

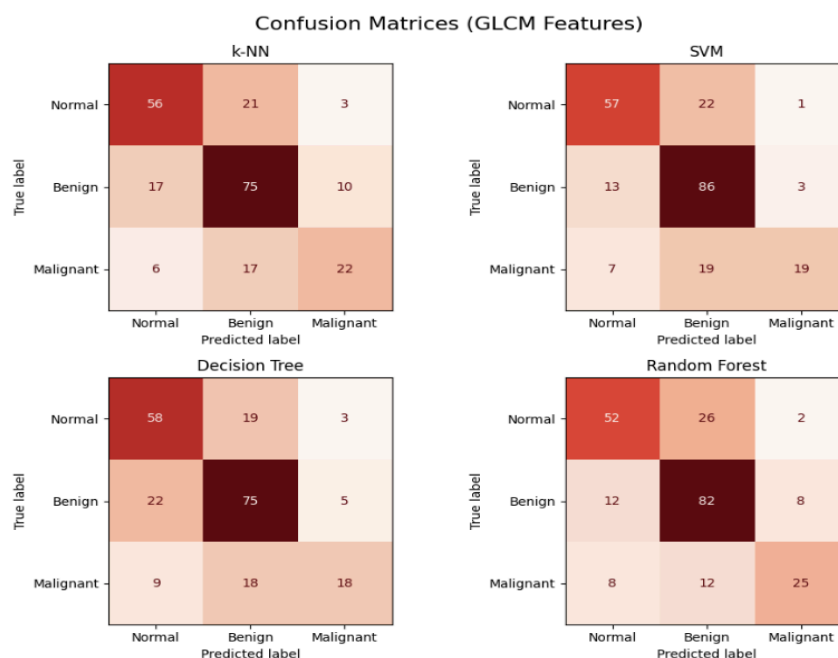


Figure 11: GLCM Confusion Matrices

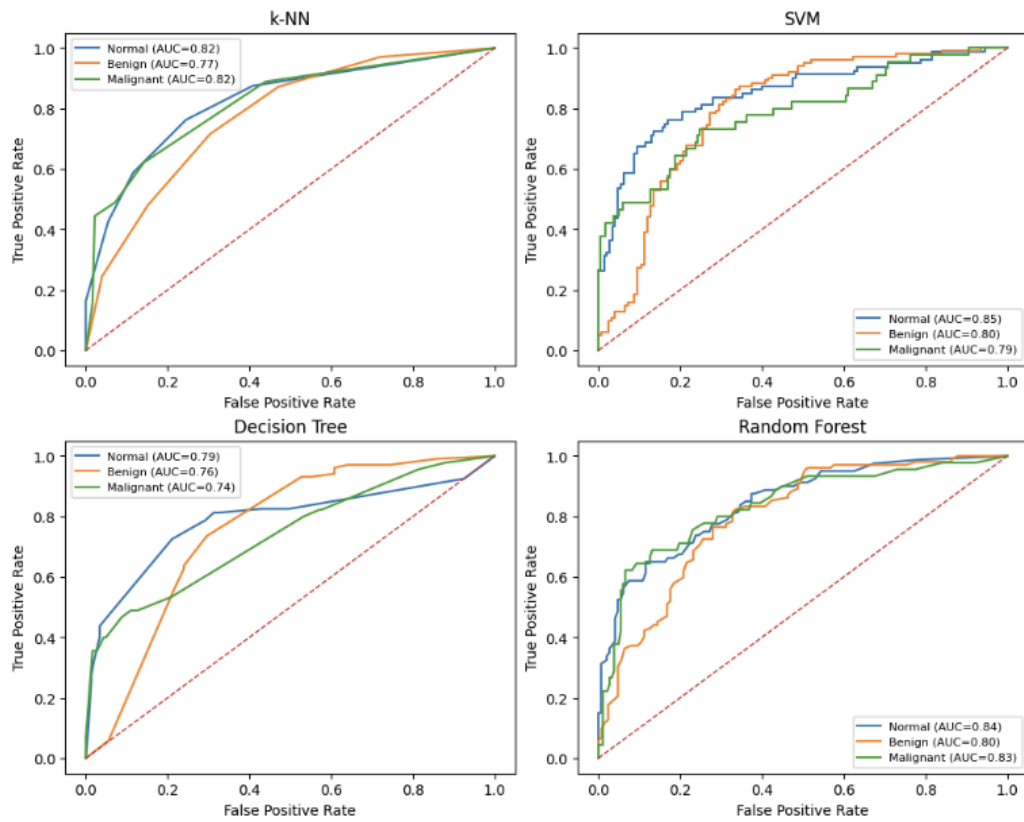


Figure 12: GLCM ROC Curves

3.3 Sprint 3

3.3.1 Implementation

The aim of sprint 3 was to implement a ResNet-18 CNN as a modern comparison to the classical machine learning approaches. (Basu et al.,2022) showed that deep learning can achieve strong predictive performance on gallbladder ultrasound images, making it important to include in this study.

A dataset class was implemented in PyTorch to load the ROI-cropped ultrasound images and their class labels.

```
class GallbladderDataset(Dataset):
    def __init__(self, labels_dict, bbox_data, img_dir, transform=None):
        self.items = []
        self.bbox_data = bbox_data
        self.img_dir = img_dir
        self.transform = transform

        for img_name, lbl in labels_dict.items():
            if img_name not in bbox_data:
                continue

            img_path = os.path.join(img_dir, img_name)
            if not os.path.exists(img_path):
                continue

            self.items.append((img_name, lbl))
```

Figure 13: Class to load ROI cropped images, ResNet-18

PyTorch was chosen for the CNN implementation because it is well-suited to transfer learning, which was important for this project because a pretrained model could be adapted rather than training a CNN from scratch. The syntax is a lot more intuitive than frameworks like TensorFlow. It provided libraries like Torchvision, which made making the dataset class much simpler.

The ROI extraction was handled using OpenCV, but the cropped image was resized to match the CNN input size.

```
img = cv2.imread(img_path)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

bbox = self.bbox_data[img_name]["bbs"][0][1]
x1, y1, x2, y2 = bbox

x1, y1 = max(0, x1), max(0, y1)
x2, y2 = min(img.shape[1], x2), min(img.shape[0], y2)

roi = img[y1:y2, x1:x2]
```

Figure 14: Crop and resize ROI

The model itself was implemented using ResNet-18 from torchvision. Because it is pretrained and well tested, it is more suited for smaller datasets than other deep learning approaches, which require more data. It was also chosen because it is a standard CNN architecture that is deep enough to learn useful visual features, but lightweight enough to remain practical on modest hardware. Using a pretrained model also made implementation more realistic for the scope of the project, since transfer learning is more suitable than training a large CNN entirely from scratch on a limited medical dataset.

The CNN was then trained using increasing numbers of epochs

```
train_losses = []

for epoch in range(EPOCHS):
    loss = train_one_epoch(model, train_loader, criterion, optimizer, device)
    train_losses.append(loss)
    print(f"Epoch {epoch+1}/{EPOCHS} - Loss: {loss:.4f}")
```

Figure 15: CNN training for N epochs

This implementation choice was important because it allowed the project to investigate how training duration affected performance, rather than treating the epoch count as arbitrary. (Goodfellow, Bengio and Courville, 2016) showed that increasing the number of training epochs allows a neural network to better learn patterns in data.

3.3.2 Validation

Validation of the CNN pipeline worked exactly like in sprint 2, using the same metrics to compare the models accurately. In addition to this, the CNN validation included monitoring the training loss each epoch to validate the actual training process.

```
plt.figure(figsize=(6, 4))
plt.plot(range(1, EPOCHS + 1), train_losses, marker='o')
plt.xlabel("Epoch")
plt.ylabel("Training Loss")
plt.title("ResNet-18 Training Loss")
plt.show()
```

Figure 16: Code to create a plot of training loss across epochs

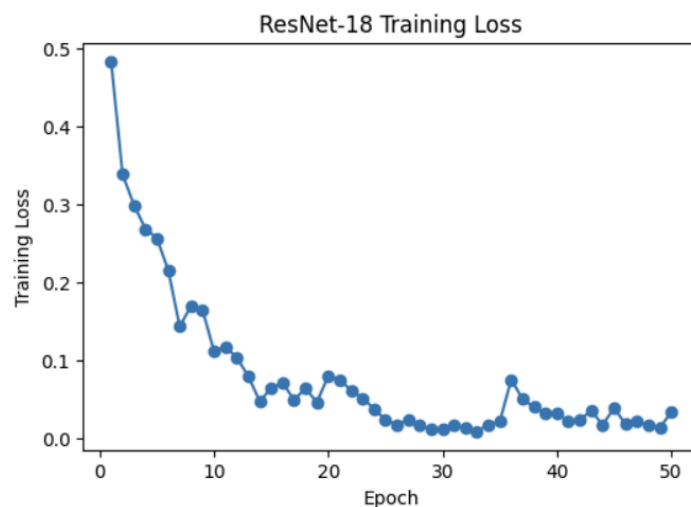


Figure 17: Graph of training loss up to 50 epochs, ResNet-18

3.4 Overall Software Validation

Across all the code files, validation checks were done. They check if labels matched image file names as well as, missing images or invalid ROI crops were skipped to avoid errors and so they don't enter the pipeline. ROI coordinates were clipped to image boundaries to avoid errors when cropping. Doing this avoided unreliable feature extraction, training and results.

```
if img is None:
    return None

# Keep roi in image bounds
x1, y1, x2, y2 = bbox
x1, y1 = max(0, x1), max(0, y1)
x2, y2 = min(img.shape[1], x2), min(img.shape[0], y2)
```

Figure 18: Validation for missing images and ROI boundary clipping

Chapter 4

Results, Evaluation and Discussion

The full set of Machine Learning Test Results, including ROC curves and confusion matrices can be found in (**Appendix C**)

4.1 Comparison of Classical Machine-Learning Approaches

4.1.1 Results

Sprint 2 formed the main classical machine-learning evaluation stage of the project. The purpose of this sprint was to compare the performance of the four selected classifiers, k-NN, SVM, Decision Tree, and Random Forest, across the three feature extraction techniques, GLCM, LBP, and PCA. This allowed the project to determine which combination of feature extraction and classifier was most effective for classifying gallbladder ultrasound images into normal, benign, and malignant categories.

Overall, PCA produced the strongest classical results, GLCM gave moderate but relatively balanced performance, and LBP was the weakest overall. The full metric comparison is summarised in **Table 1**.

Feature extraction	Model	Accuracy	Precision	Recall	F1-score
GLCM	k-NN	0.6740	0.6671	0.6414	0.6507
GLCM	SVM	0.7137	0.7478	0.6593	0.6787
GLCM	Decision Tree	0.6652	0.6712	0.6201	0.6315
GLCM	Random Forest	0.7004	0.7066	0.6698	0.6827
LBP	k-NN	0.6872	0.6035	0.5959	0.5853
LBP	SVM	0.7137	0.4740	0.5887	0.5249

LBP	Decision Tree	0.6960	0.5448	0.5812	0.5423
LBP	Random Forest	0.7093	0.6415	0.6173	0.6101
PCA	k-NN	0.6916	0.6874	0.6518	0.6572
PCA	SVM	0.7930	0.7471	0.7127	0.7171
PCA	Decision Tree	0.7445	0.7135	0.7187	0.7131
PCA	Random Forest	0.7841	0.7640	0.6901	0.6914

Table 2: Metric results for all models

The strongest overall classical pipeline was SVM with PCA, which achieved the highest accuracy (0.7930) and the highest F1-score (0.7171). The next strongest classical result was Random Forest with PCA, which achieved accuracy (0.7841) and the highest precision (0.7640) among the PCA-based models. Decision Tree with PCA also performed well, particularly in terms of recall (0.7187), suggesting that PCA was the most effective at separating the different classes. Meaning that, for this dataset, it managed to reduce the dimensionality while still preserving important variance patterns in the input space.

By contrast, the weakest classical performance was observed with LBP, particularly when used with SVM. Although LBP-SVM achieved an accuracy of 0.7137, its precision and F1-score were substantially lower than the corresponding GLCM and PCA pipelines. This shows that accuracy alone would have been misleading, and supports the decision to use an evaluation set including precision, recall, F1-score, confusion matrices, and ROC curves.

A clear pattern across all three feature extraction methods was that benign was generally the easiest class to identify, while malignant was the hardest. Sarangi *et al.* (2025) note that ultrasound does not reliably distinguish benign from malignant gallbladder wall thickening, while Basu *et al.* (2022) argue that malignant gallbladder cases are especially difficult because the gallbladder boundary and shape are often unclear due to noise. The confusion matrices in this study strongly support this, as malignant cases were repeatedly misclassified as benign across multiple pipelines.

4.1.2 GLCM Results and Discussion

The GLCM results showed that this texture-based representation was a reasonable baseline, but not the strongest feature extraction method overall. Within the GLCM pipelines, Random Forest achieved the highest F1-score (0.6827) and recall (0.6698), while SVM achieved the highest accuracy (0.7137) and precision (0.7478).

The confusion matrices indicate that benign was the easiest class to identify under GLCM, while malignant remained the most difficult. For example, with GLCM-SVM, 86 benign cases were correctly classified, compared with only 19 malignant cases. Similar patterns were visible across the other GLCM models. This suggests that although GLCM captured useful statistical texture information, it was not enough to separate malignant abnormalities reliably from benign ones. This is supported by the presentation of malignant and benign cases being very similar in ultrasound images, occurring as slight wall thickening and small tumours, as evidenced in (section 1.2.1)

The ROC curve results support this interpretation. Using GLCM, the best Normal Area Under the Curve (AUC) was achieved by SVM (0.85), while the best Malignant AUC was achieved by Random Forest (0.83). These values indicate good class separability, but they were still weaker than the best PCA-based results.

4.1.3 LBP Results and Discussion

The LBP results were generally weaker than both GLCM and PCA. The strongest LBP model was Random Forest, with accuracy (0.7093), precision (0.6415), recall (0.6173), and F1-score (0.6101). Although SVM matched the GLCM-SVM accuracy at 0.7137, its overall behaviour was poor when examined using the confusion matrix and F1-score.

The confusion matrix for LBP-SVM reveals that it predicted zero malignant cases correctly. All malignant cases were classified as Benign, resulting in an artificially acceptable overall accuracy but poor precision and F1-score. This is an important result because it shows why multiple evaluation metrics were necessary. If accuracy had been considered alone, the performance of LBP-SVM would have appeared much stronger than it actually was. This showed that LBP-SVM struggled to differentiate between benign and malignant classes.

The ROC curves also show weak separation of the malignant class and other classes for LBP. The best malignant AUC under LBP was Random Forest (0.73), while Decision Tree achieved only 0.59 for the malignant class. A likely explanation is that LBP focuses on very local texture patterns, whereas ultrasound malignancy may also depend on broader textural and structural relationships to separate benign and malignant cases accurately.

The LBP results suggest that local micro-texture alone was not the most effective representation for this dataset. It provided useful information, but it was less reliable than both GLCM and PCA, particularly for the malignant class.

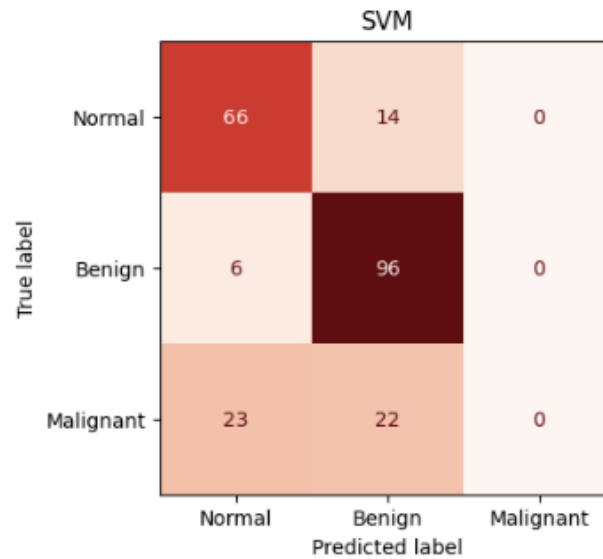


Figure 19: LBP-SVM confusion matrix, 0 malignant cases identified

4.1.4 PCA Results and Discussion

The PCA results were the strongest among the classical approaches. All four classifiers improved relative to their GLCM and LBP versions, particularly for SVM and Decision Tree.

The best overall classical result was PCA-SVM, which achieved accuracy 0.7930, precision 0.7471, recall 0.7127, and F1-score 0.7171. PCA-Random Forest also performed strongly, with accuracy 0.7841, while PCA-Decision Tree achieved the highest recall among the classical pipelines at 0.7187. These results show that PCA provided a cleaner and more separable feature space for the classifiers than texture-based feature extraction techniques. Especially because PCA improved performance across all four model types, implying that the improvement came from the feature extraction method rather than from one specific classifier.

The ROC curves support this interpretation. PCA-SVM achieved AUC = 0.99 for the Normal class, 0.88 for Benign, and 0.80 for Malignant, while PCA-Random Forest achieved 0.98, 0.88, and 0.82, respectively. These were the strongest ROC results among the classical feature extraction methods. Even though malignant classification remained more difficult than the other two classes, the ROC results show PCA clearly improved class separability more than GLCM and LBP.

This is an important result because it shows that, for this dataset, compressing the ROI into principal components was more useful than relying only on handcrafted texture data.

4.2 ResNet-18 CNN

4.2.1 Quantitative Results

Sprint 3 implemented and evaluated a ResNet-18 CNN as the project's modern deep-learning approach. The model was trained and evaluated using 20, 50, and 100 epochs in order to investigate how training duration affected performance.

Epochs	Accuracy	Precision	Recall	F1-score
20	0.7131	0.7576	0.7268	0.7280
50	0.7787	0.8525	0.7823	0.7822
100	0.7459	0.7729	0.7653	0.7610

Table 3: ResNet-18 CNN metric results

These results show that the CNN improved substantially from 20 to 50 epochs, but then declined at 100 epochs. The best-performing configuration was therefore 50 epochs, with the highest accuracy (0.7787), precision (0.8525), recall (0.7823), and F1-score (0.7822).

This is an important finding because it shows that increasing the number of epochs was beneficial up to a point, but further training did not continue to improve performance. Instead, the drop from 50 to 100 epochs suggests diminishing returns and possible overfitting.

4.2.2 Confusion Matrix and ROC Discussion

At 20 epochs, the model performed strongly on the Normal class, correctly classifying 29 out of 31 Normal images, but was weaker on Benign and Malignant. In particular, 24 malignant cases were misclassified as benign, indicating that the CNN had not yet learned sufficient separation between the 2 classes. The ROC curves reflect this, with AUC = 1.00 for Normal, but only 0.77 for Benign and 0.70 for Malignant.

At 50 epochs, the confusion matrix showed the best overall behaviour. Normal remained very strong, Benign improved substantially, and malignant classification also improved compared with 20 epochs. However, distinguishing between malignant and benign classes still proved to be difficult, with 23 misclassified malignant cases. The ROC curves were also strongest at this stage, with AUC = 1.00 for Normal, 0.85 for Benign, and 0.81 for Malignant.

At 100 epochs, Normal classification became perfect in the confusion matrix, with 31/31 Normal cases correctly classified, but Benign and Malignant performance did not improve accordingly. Malignant classification remained difficult, with 21 malignant cases misclassified as benign, and the overall accuracy fell below the 50-epoch result. The ROC curves also weakened compared with 50 epochs, particularly for Benign and Malignant. This suggests that the additional training was not improving generalisation.

4.2.3 Training Loss and Overfitting Discussion

Training loss is the error between the predicted outputs and the actual targets (Bergmann and Stryker, 2025). The loss decreased rapidly during the early stages of training and then remained very low for much of the remaining training process. In isolation, that might appear positive, but the decline in test performance from 50 to 100 epochs shows that better fitting of the training set was not translating into better performance on unseen data beyond a certain point. This pattern is consistent with mild overfitting, where continued improvement on the training data does not translate into improved performance on test data (Bartlett *et al.*, 2023). These findings are consistent with what was discussed in **(section 2.2.5)** about the potential of overfitting when using a large number of epochs on smaller datasets.

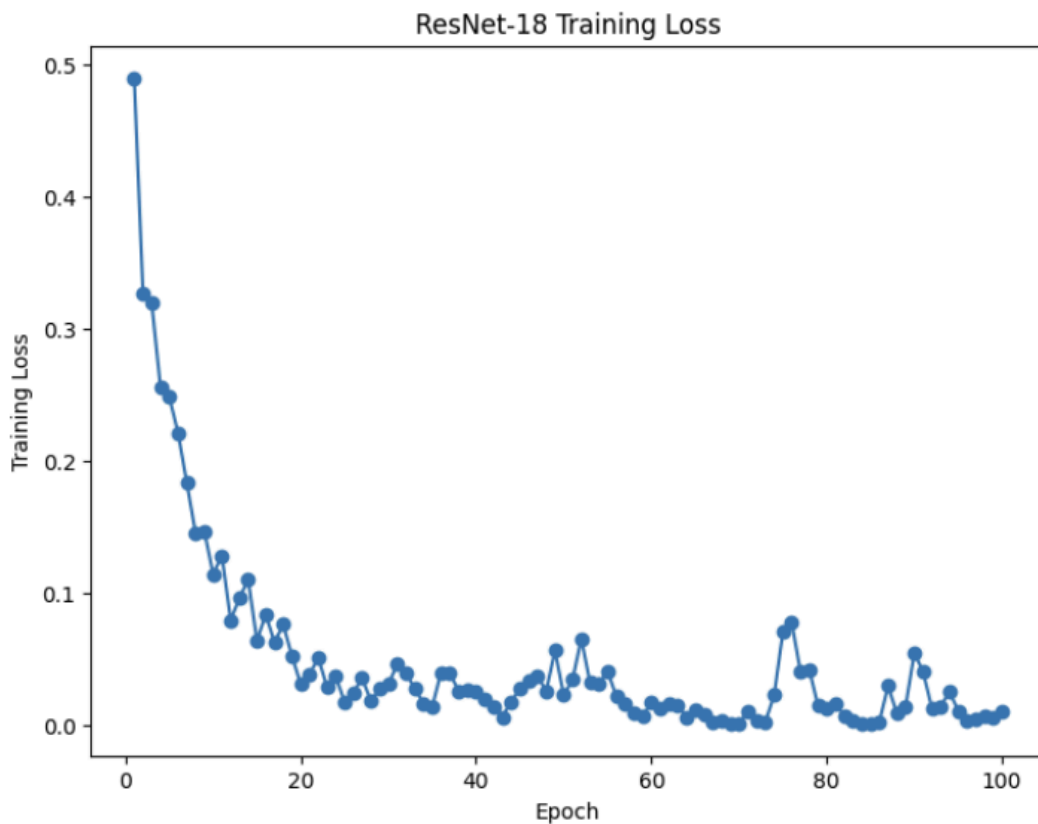


Figure 20: Training loss up to 100 epochs, ResNet-18 CNN

5.1 Evaluation and Conclusions

The overall aim of the project was to investigate whether machine-learning methods could improve the classification of gallbladder ultrasound images and to compare classical feature-based approaches with a modern CNN-based approach. The results show that this aim was achieved, and that the main objectives of the project were also met.

The project successfully preprocessed the ultrasound images and used the region-of-interest annotations to focus analysis on the gallbladder region. This was an important objective because it ensured that the models learned from the diagnostically relevant part of the image rather than from irrelevant background information.

The project successfully implemented GLCM, LBP, and PCA as feature extraction strategies. The results from Sprint 2 showed that these three representations produced clearly different outcomes, demonstrating that feature extraction had a significant impact on classification performance. PCA was the strongest overall, outperforming both GLCM and LBP across the classical models.

The project met the objective to train and evaluate k-NN, SVM, Decision Tree, and Random Forest classifiers in combination with the feature extraction techniques. This comparison showed that classifier choice was also important, with SVM and Random Forest generally performing better than k-NN and Decision Tree.

The project met the objective to implement a ResNet-18 CNN as a modern comparison model. Sprint 3 showed that its performance varied depending on the number of training epochs used. The best CNN result was achieved at 50 epochs, showing that the network benefited from longer training up to a point, but did not continue improving at 100 epochs.

5.1.1 Classical Models versus ResNet-18

- Best classical accuracy: PCA-SVM (0.7930)
- Best classical F1-score: PCA-SVM (0.7171)
- Best CNN accuracy: ResNet-18 at 50 epochs (0.7787)
- Best CNN F1-score: ResNet-18 at 50 epochs (0.7822)

The best classical model, PCA-SVM, was slightly better in terms of overall accuracy. However, in a medical context, where false positives and false negatives can have serious consequences, the stronger F1-score of the CNN is arguably more important. At the same

time, the fact that PCA-SVM remained highly competitive supports the view that classical pipelines can still be very effective on moderate-sized, noisy ultrasound datasets.

For both classical and deep learning models, the confusion matrices repeatedly showed malignant cases being assigned to the benign class. This aligns closely with medical literature, such as (Rana et al., 2024), showing that ultrasound often cannot reliably distinguish benign from malignant wall thickening. The project results, therefore, reflect a genuine challenge of ultrasound imaging rather than a weakness of the models themselves.

The difficulty distinguishing malignant from benign cases, greatly reduced the accuracy scored of the models. To conclude, machine learning classification can be successfully applied to ultra sound images, seen by the models strong differentiation of normal and benign cases. However, in the context of gall bladder cancer, reliably detecting malignancy is poor with the current dataset size. A larger dataset could allow a model to better distinguish between benign and malignant cases.

5.1.1 Limitations and Future Work

Although the project reached its objectives and main aims, several limitations should be acknowledged.

The GBCU dataset, with 1255 images, is modest compared to the much larger datasets used to train CNNs. This likely constrained the CNN and led to the overfitting seen at higher epoch counts, evidenced in **(section 4.2.3)**. Future work could involve using a larger dataset to improve the performance of CNNs and allow me to use deeper models like ResNet-50, which have better performance (He et al.,2016).

The project relied on the radiologist-provided ROI annotations. This means classification was made easier through localisation. An automated clinical system would need ROI detection capabilities or perform well without them. Future work could be building such a system.

No tuning was done to the classical machine learning methods themselves. For example, experimenting with different k values for K-NN or different depths for decision trees. Results were based on the specific configuration used. Therefore, some models may have done better with more extensive optimisations. Future work could involve a study into different configurations of the machine learning models.

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Appendix A Self-appraisal

A.1 Critical self-evaluation

I believe the project achieved its main aim of investigating and comparing classical machine learning and deep learning approaches for gallbladder ultrasound classification. Developing a complete pipeline and experimenting with different feature extraction techniques, classical machine learning models and a ResNet-18 CNN. The results showed clear differences between the different approaches, finding that for the GBCU dataset, PCA-SVM and 50 epochs for Res-Net18 achieved the best results. Furthermore, because ResNet18 produced a higher F1-score, it was more suited to be used in a clinical setting. Therefore, the project produced a meaningful comparison and evaluation.

Another strong aspect of the project was the evaluation's structure. Using a standardised pipeline for the classical machine learning models made their results more reliable. Using a combination of different metrics (precision, recall, F1-score, confusion matrices and ROC curves) and not accuracy alone was also a strong choice. This was important because some approaches, such as LBP-SVM, appeared acceptable based on accuracy alone but performed poorly when examined more closely with other metrics.

Another strength was how strongly the results linked to established medical literature, which further validated their reliability. The Background research found that benign and malignant gallbladder abnormalities can present very similarly and that they were difficult to distinguish using ultrasound imaging. The results, in particular the confusion matrices, reflected this challenge.

However, there were a few limitations in the project methodology (**section 5.1.1**). One of which is the limited dataset size. The GBCU dataset is modest compared to the much larger datasets used in deep learning. This likely constrained the CNN's performance and led to the overfitting seen in higher epoch counts. A larger dataset could also have enabled the use of deeper models, such as ResNet-50, and therefore achieved better performance during evaluation.

Another limitation is the lack of tuning done on the classical machine learning models themselves. For example, experimenting with different k values of K-NN or different tree depths for decision trees could have improved their performance. The evaluations reflected a specific, potentially limiting, configuration of the models rather than their maximum possible performance.

A.2 Personal reflection and lessons learned

One of the most important lessons I learned was that model performance in medical imaging depends not only on the classifier but also heavily on the feature extraction and preprocessing used on the data. Before experimentation, I expected the main differences would come from the classifiers; however, the results showed that changing between GLCM, LBP and PCA had a much bigger effect on performance.

I also learned the importance of using a combination of metrics when evaluating a model's performance. Using accuracy alone doesn't provide a complete picture of a model's characteristics. A key example was the LBP-SVM results (**Appendix C**), which appeared to have a good accuracy, but the confusion matrices revealed that no malignant cases were correctly identified. I personally now understand the importance of metrics like precision, recall and F1-Score, instead of viewing them as arbitrary numbers.

The incorporation of a CNN was a valuable learning experience and my first time using deep learning. It improved my understanding of PyTorch and how residual networks work. The epoch comparison was especially useful, because it showed how longer training can improve performance up to a point, but may then reduce generalisation. This improved my conceptual understanding of overfitting.

I learned a lot about ultrasound imaging and gallbladder conditions throughout the project. It also made me more confident in using machine learning, both modern and classical, for medical image analysis.

I learned I need to have more confidence in my own ability. During the development process, I felt that I was not capable of performing this study in the time I had. This was due to my not having a complete understanding of machine learning, which led me to overcomplicate the development of the models. This slowed me down and restricted my aims for this project. I could have achieved more with my current understanding of machine learning, and in particular, deep learning.

A.3 Legal, social, ethical and professional issues

A.3.1 Legal issues

The main legal issue in this project was the use of medical imaging data. Although the GBCU dataset is publicly released for research purposes and describes the images as anonymised, medical data is still inherently sensitive. Therefore, I only used the dataset within the scope of an academic study, which is why it was released. The dataset site also states that written consent from patients was obtained.

Another legal issue is licensing concerns. The project relied on external material, including the GBCU dataset as well as multiple Python libraries. These resources were used in accordance with their licenses and credited in the report.

Despite this project being a study and the models not being intended for clinical use, if such a system were developed, then questions about liability when the model misdiagnosed a patient would arise.

A.3.2 Social issues

The project addresses the socially important issue of improving the analysis of gallbladder ultrasound images. If machine learning models were improved to a point where they could be used to help support diagnosis, it could lead to more cases of cancer being caught at an early stage, making the disease itself more treatable and helping many people.

At the same time, a system like this could instil false confidence if users assumed it was always accurate. This is particularly problematic because the results showed how easily models confused benign and malignant cases. This could lead to misplaced reassurance or unnecessary concern, both of which are bad.

The GBCU dataset was created using data from a hospital in Chandigarh, North India. This is a very specific clinical context, and therefore, the models may not generalise equally well across other hospitals, patient groups and ultrasound machines. A more refined model could use a larger variety of ultrasound image data.

A.3.3 Ethical issues

Transparency is an ethical concern. Classical models with handcrafted features create predictions that are understandable. However, deep learning and neural networks are considered a “black box”, meaning the internal decision-making used to reach a prediction is not easily understood. This makes them harder to trust by patients in a clinical setting.

Despite this only being a research study, the use of AI in a medical context has ethical risks. Errors can have negative consequences, and who is to be held responsible can be unclear. This is highlighted by the study showing that malignant cases were often confused with benign ones. Therefore, a system like this should be viewed, at most, as a support tool rather than a replacement for a diagnostician.

Collecting personal patient data is also an ethical issue. Informed written consent from the patients was obtained when making the dataset by the team at IIT Delhi.

A.3.4 Professional issues

I followed professional standards in report writing and technical work. I structured the project into sprints, following a well-established and widely used development methodology. I used GitHub for version control. I properly credited external sources, which I used to develop the project, in my report. These included libraries, pretrained models and published literature. My code adhered to the PEP 8 coding standard for Python, making it more readable and consistent.

Another issue is the responsible presentation and interpretation of results. I showed results for all metrics across all models, not just the strongest-performing metric values. It was important to discuss weaker behaviour, such as the models' confusing benign and malignant cases.

Appendix B

External Materials

The dataset used:

GBCU dataset – Gallbladder Cancer Ultrasound dataset: <https://gbc-iitd.github.io/data/gbcu>

Which accompanies the paper “Surpassing the Human Accuracy: Gallbladder Cancer Detection from USG with Curriculum Learning,” *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 20854–20864, by Basu, S., Gupta, M., Rana, P., Gupta, P. and Arora, C.

Pretrained model:

ResNet-18 (pretrained) – used through torchvision: <https://pytorch.org/vision/stable/>

Libraries and Frameworks:

scikit-image – GLCM and LBP feature extraction: <https://scikit-image.org/docs/stable/>

scikit-learn – PCA, classical classifiers, and evaluation metrics:

<https://scikit-learn.org/stable/>

PyTorch – CNN training and implementation: <https://pytorch.org/docs/stable/>

Matplotlib – graphs and visualisations: <https://matplotlib.org/stable/>

Jupyter Notebook – experimentation environment: <https://jupyter.org/>

OpenCV – image loading, cropping, grayscale conversion, and resizing:

<https://github.com/opencv/opencv>

Google Colab – cloud-based environment used to train the ResNet-18 model, as it took significantly less time than if I trained it locally: <https://colab.research.google.com/>

Feature Extraction Techniques:

GLCM – scikit-image documentation:

<https://scikit-image.org/docs/stable/api/skimimage.feature.html>

LBP – scikit-image documentation:

<https://scikit-image.org/docs/stable/api/skimimage.feature.html>

PCA – scikit-learn documentation:

<https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html>

Classifiers:

k-Nearest Neighbours (k-NN) – scikit-learn documentation:

[https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.KNeighborsClassifier.htm](https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.KNeighborsClassifier.html)
!

Support Vector Machine (SVM) – scikit-learn documentation:

<https://scikit-learn.org/stable/modules/svm.html>

Decision Tree – scikit-learn documentation: <https://scikit-learn.org/stable/modules/tree.html>

Random Forest – scikit-learn documentation:

[https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.](https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html)
[html](https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html)

PEP8 Guidelines - Tutorial: <https://www.youtube.com/watch?v=wKECxtmbvYQ>

Figma - used to make pipeline diagram: <https://www.figma.com/make/>

Appendix C ML Test Results

C.1 Classical Machine Learning Results

C.1.1 GLCM Results

	Model	Accuracy	Precision	Recall	F1 Score
0	k-NN	0.674009	0.667050	0.641394	0.650692
1	SVM	0.713656	0.747837	0.659286	0.678677
2	Decision Tree	0.665198	0.671212	0.620098	0.631456
3	Random Forest	0.700441	0.706614	0.669826	0.682650

Table 4: GLCM with classical classifiers results

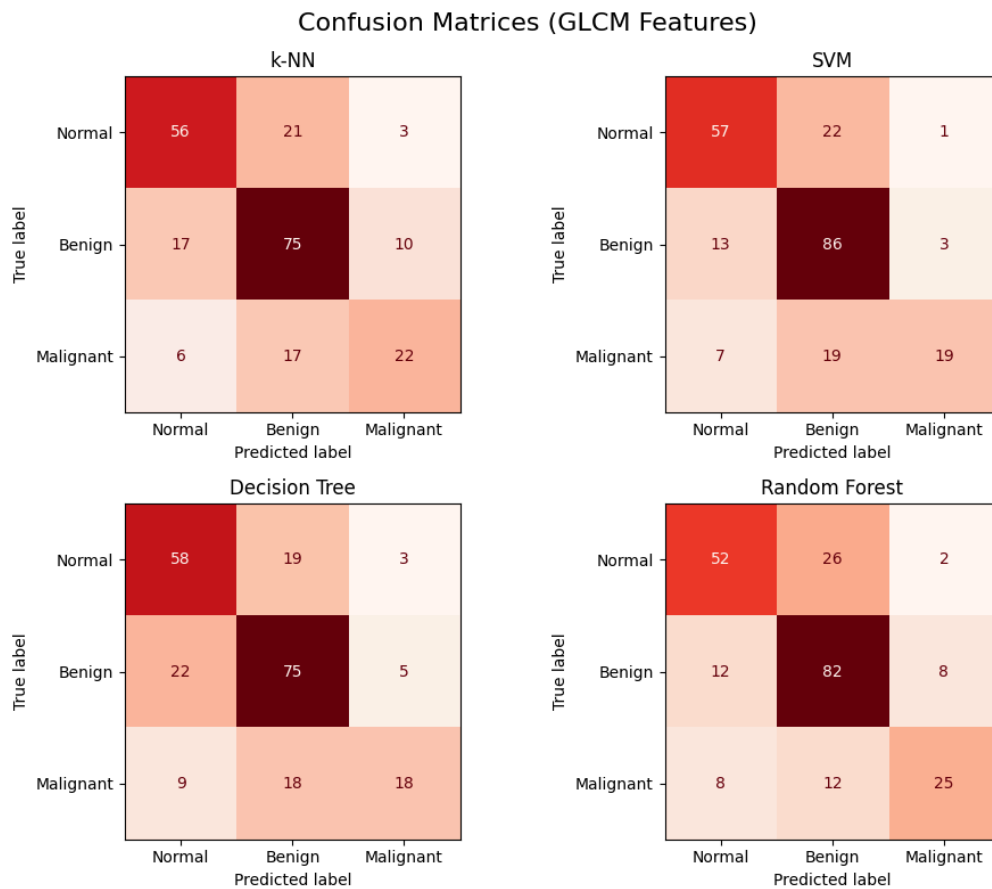


Figure 21: GLCM with classical classifiers confusion matrices

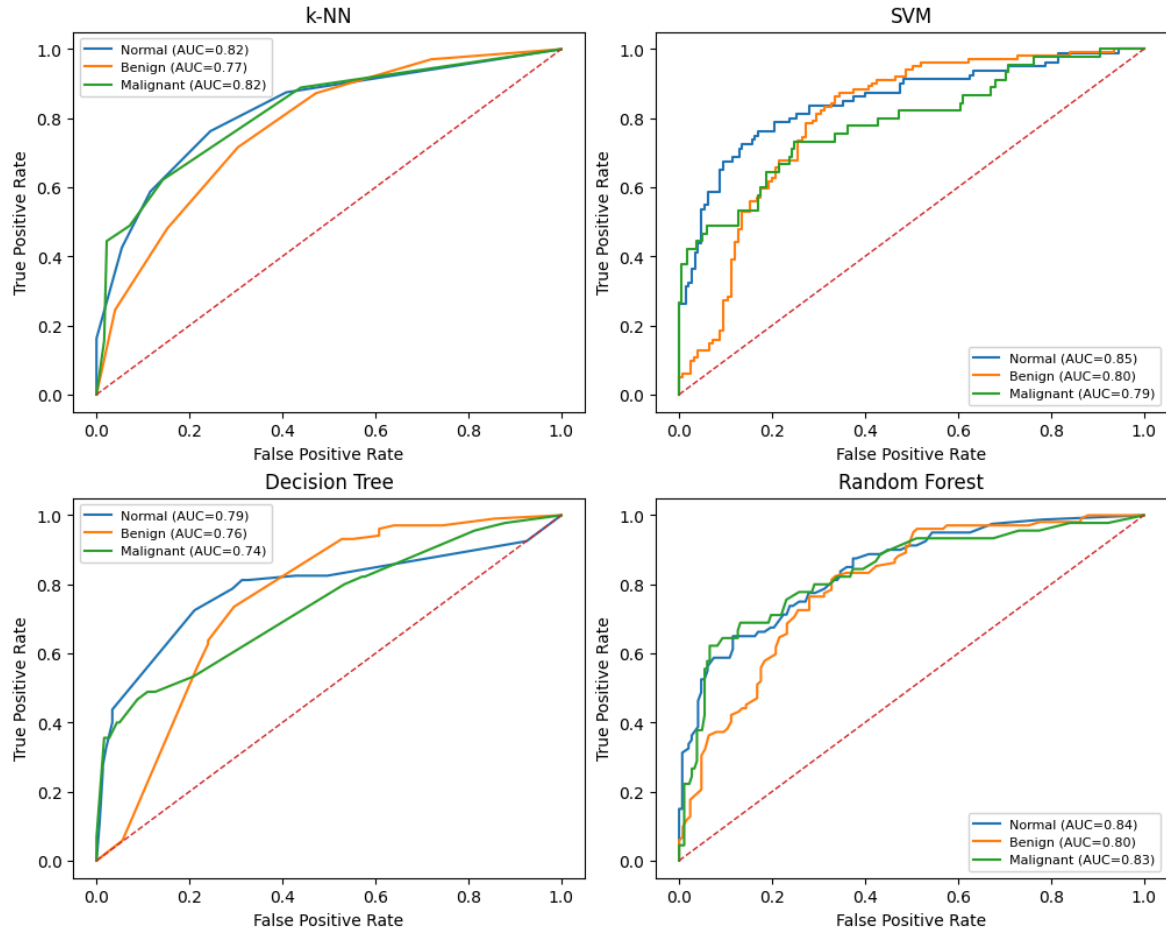


Figure 22: GLCM with classical classifiers ROC curves

C.1.2 LBP Results

	Model	Accuracy	Precision	Recall	F1 Score
0	k-NN	0.687225	0.603536	0.595942	0.585261
1	SVM	0.713656	0.474003	0.588725	0.524933
2	Decision Tree	0.696035	0.544809	0.581236	0.542328
3	Random Forest	0.709251	0.641470	0.617320	0.610139

Table 5: LBP with classical classifiers results

Confusion Matrices (LBP Features)

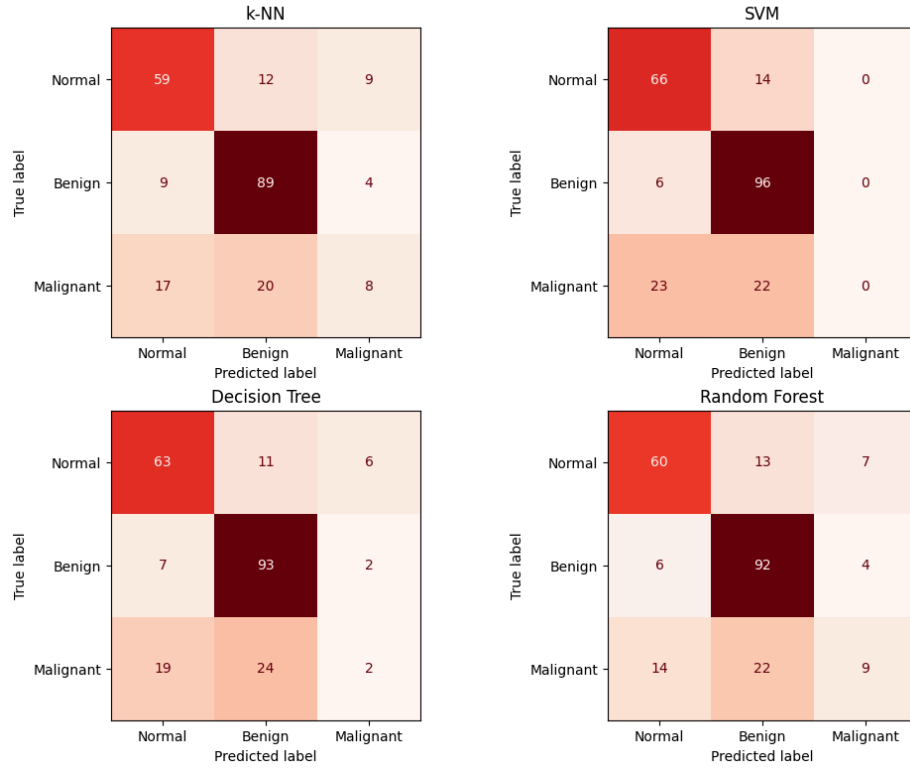


Figure 23: LBP with classical classifiers confusion matrices

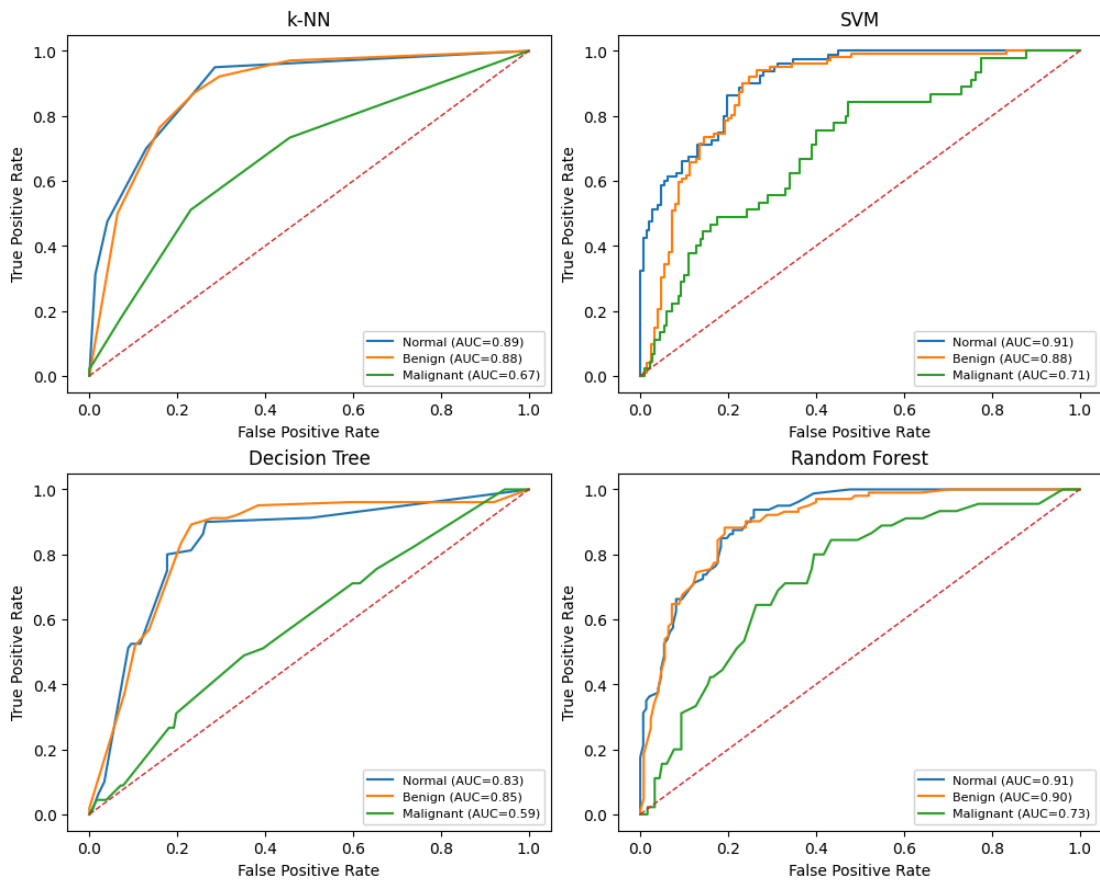


Figure 24: LBP with classical classifiers ROC curves

C.1.3 PCA Results

	Model	Accuracy	Precision	Recall	F1 Score
0	k-NN	0.691630	0.687415	0.651770	0.657210
1	SVM	0.792952	0.747145	0.712691	0.717067
2	Decision Tree	0.744493	0.713460	0.718682	0.713053
3	Random Forest	0.784141	0.763997	0.690142	0.691436

Table 6: PCA with classical classifiers results

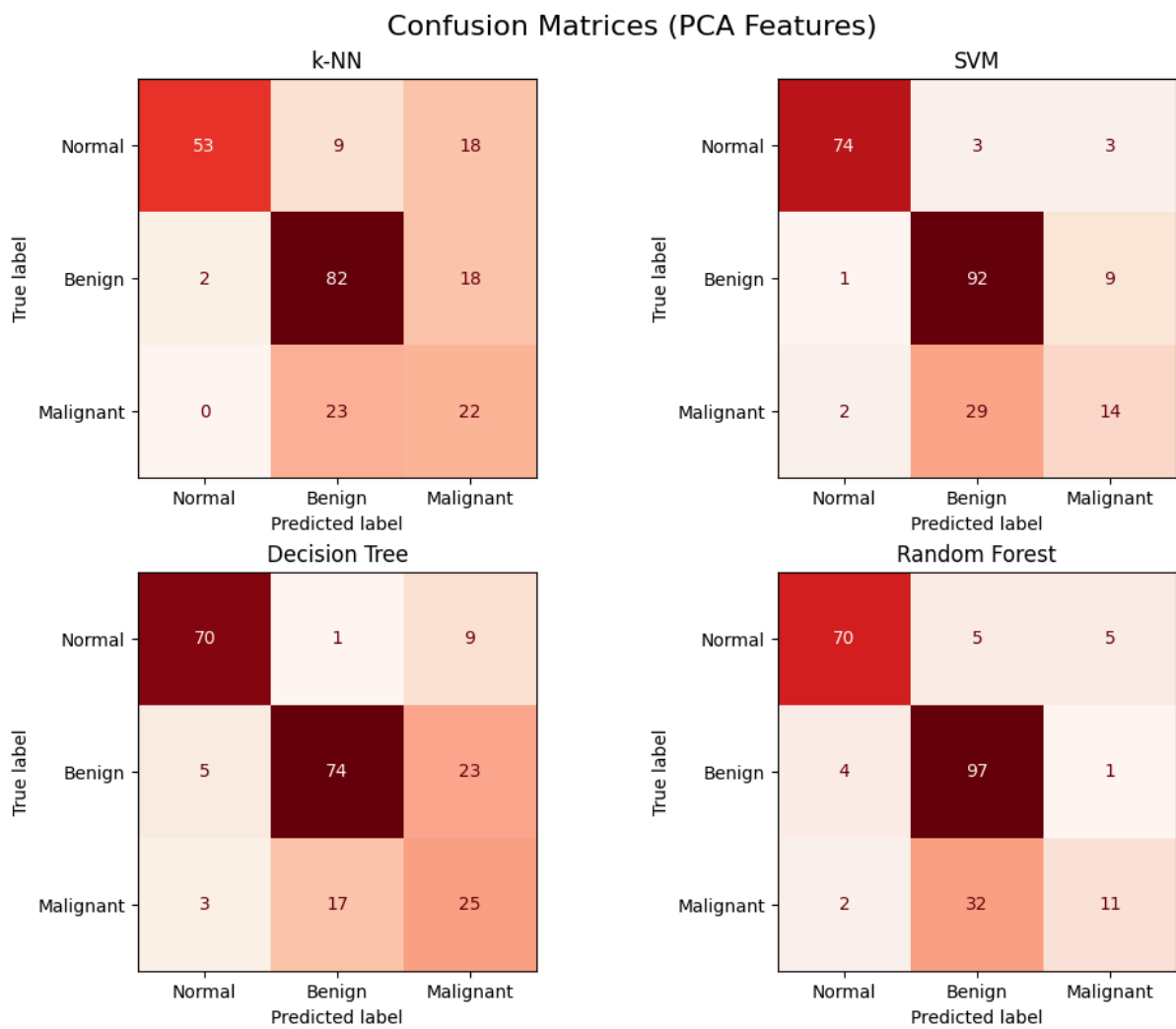


Figure 25: PCA with classical classifiers confusion matrices

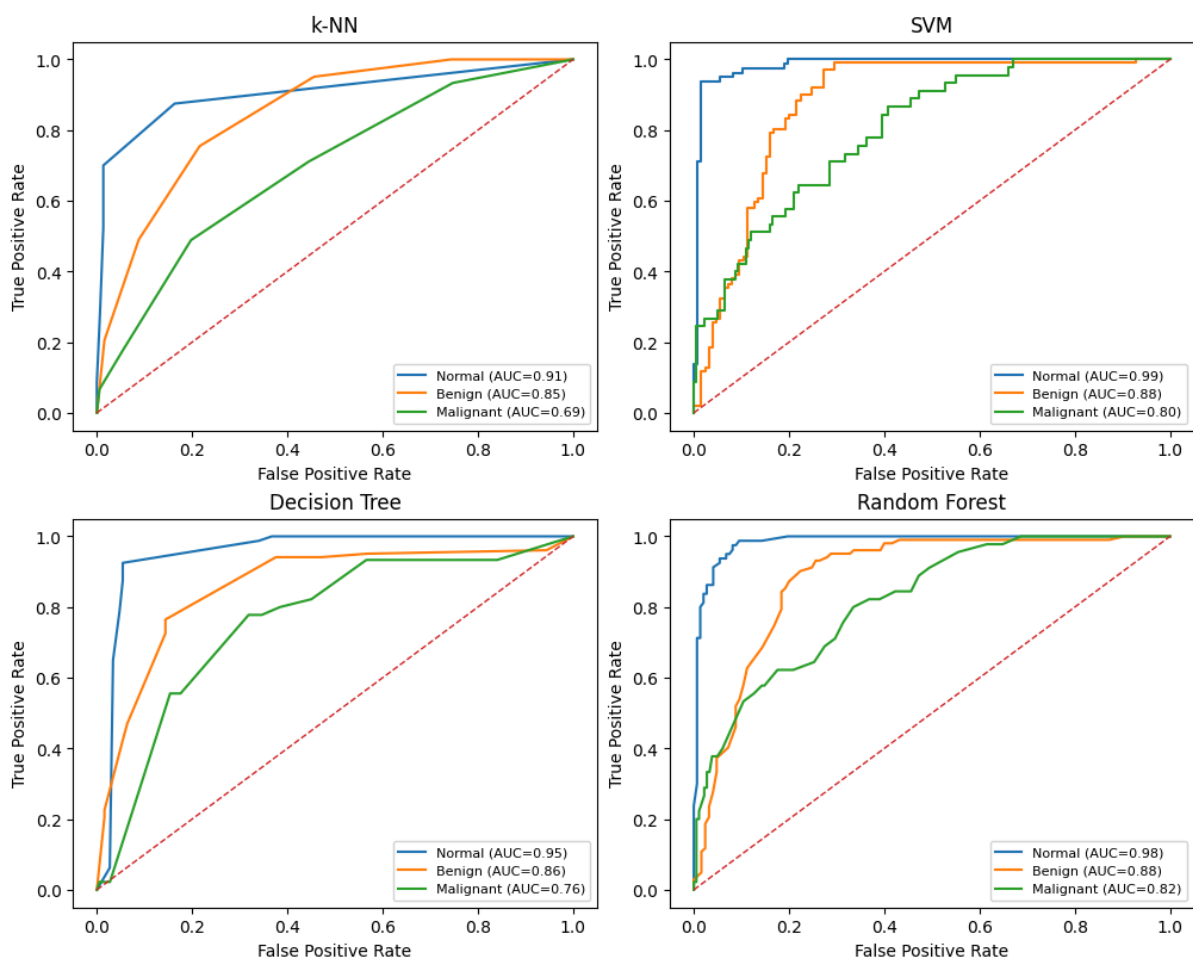


Figure 26: PCA with classical classifiers ROC curves

C.2 ResNet-18 CNN Results

C.2.1 20 Epochs Training Results

	Model	Accuracy	Precision	Recall	F1 Score
0	ResNet-18	0.713115	0.757576	0.726794	0.728019

Table 7: ResNet-18 CNN, 20 Epochs, Results

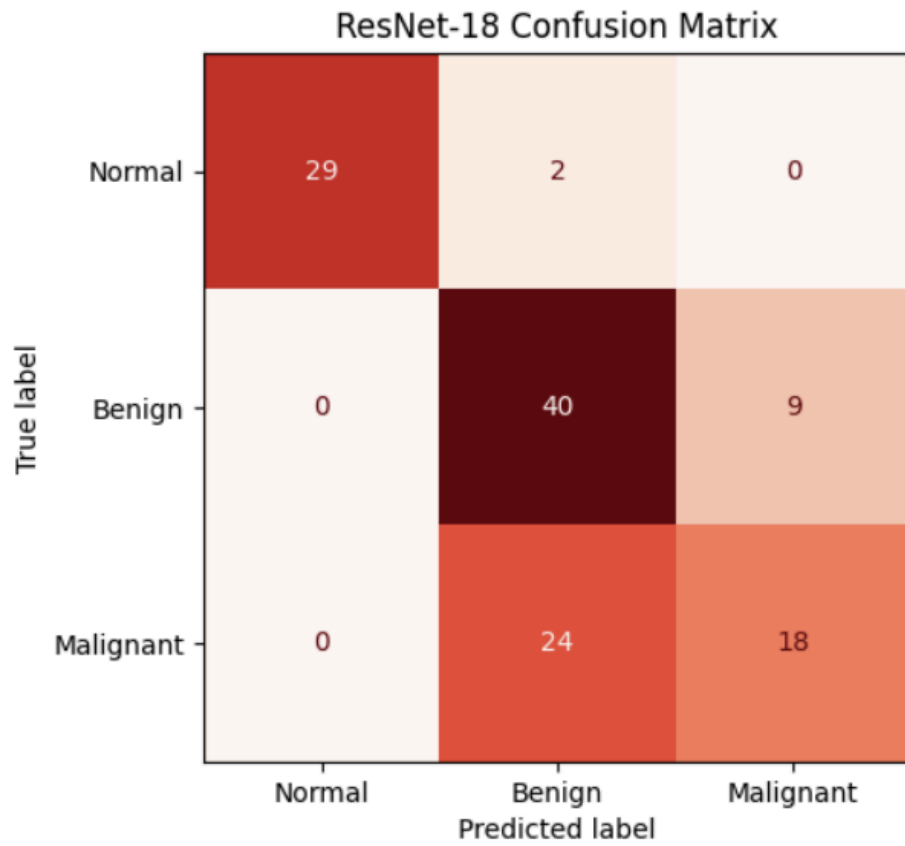


Figure 27: ResNet-18 CNN, 20 Epochs, confusion matrix

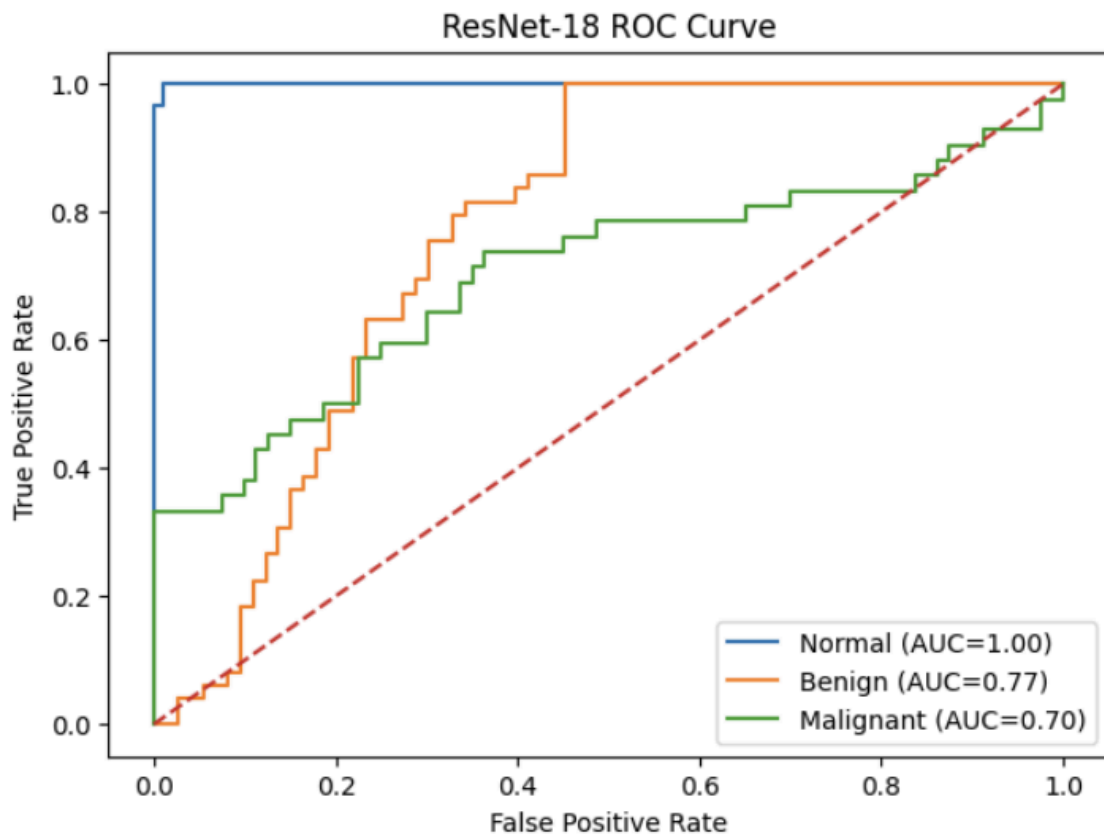


Figure 28: ResNet-18 CNN, 20 Epochs, ROC curve

C.2.2 50 Epochs Training Results

	Model	Accuracy	Precision	Recall	F1 Score
0	ResNet-18	0.778689	0.852513	0.782349	0.782234

Table 8: ResNet-18 CNN, 50 Epochs, Results

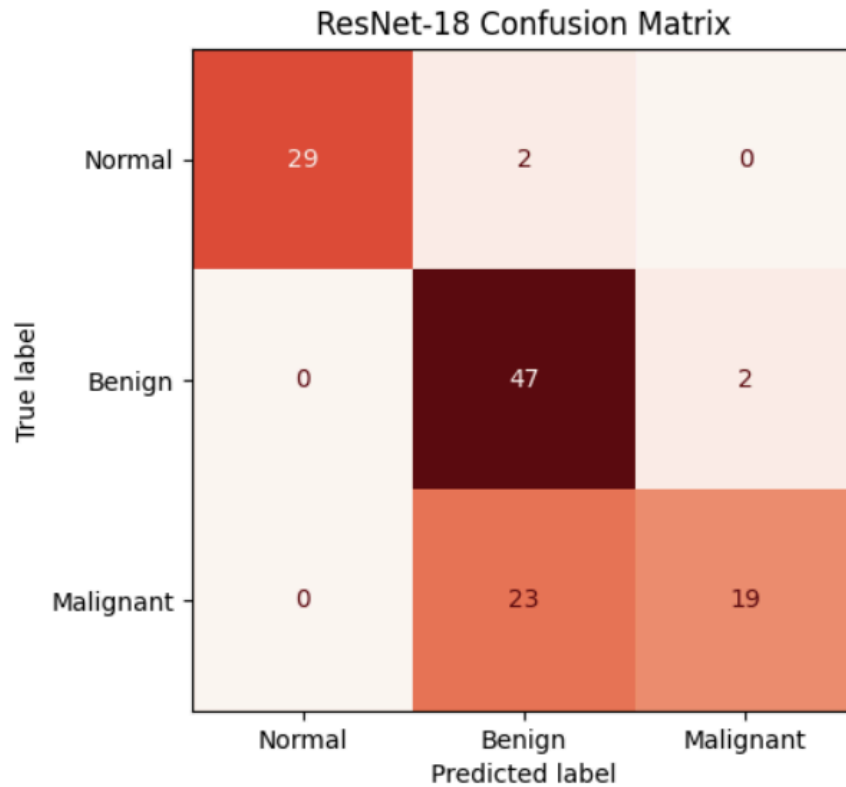


Figure 29: ResNet-18 CNN, 50 Epochs, confusion matrix

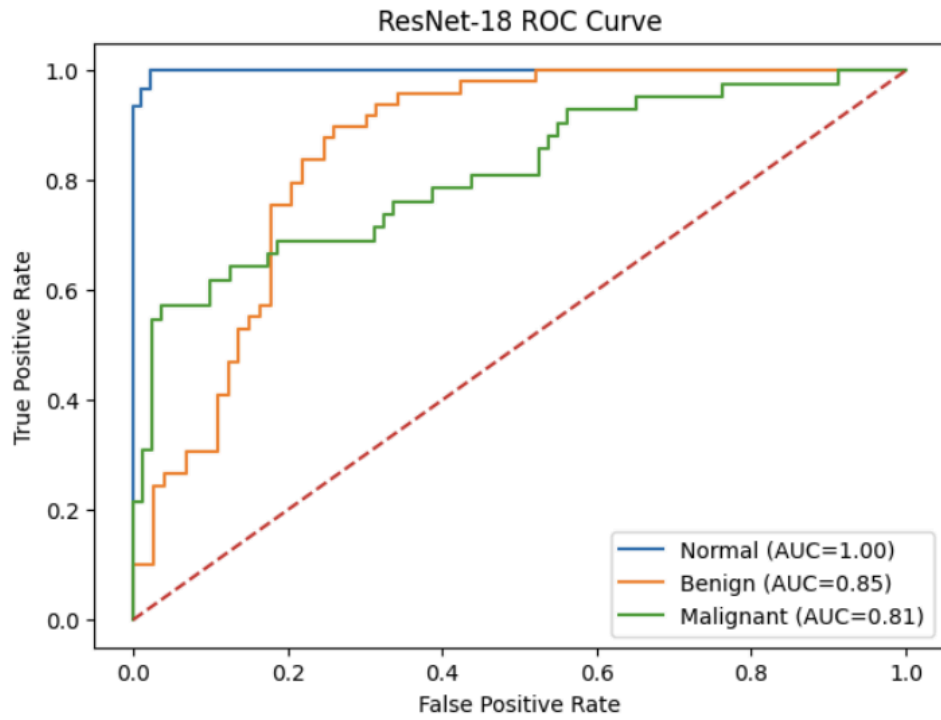


Figure 30: ResNet-18 CNN, 50 Epochs, ROC curve

C.2.3 100 Epochs Training Results

	Model	Accuracy	Precision	Recall	F1 Score
0	ResNet-18	0.745902	0.772917	0.765306	0.761019

Table 9: ResNet-18 CNN, 100 Epochs, Results

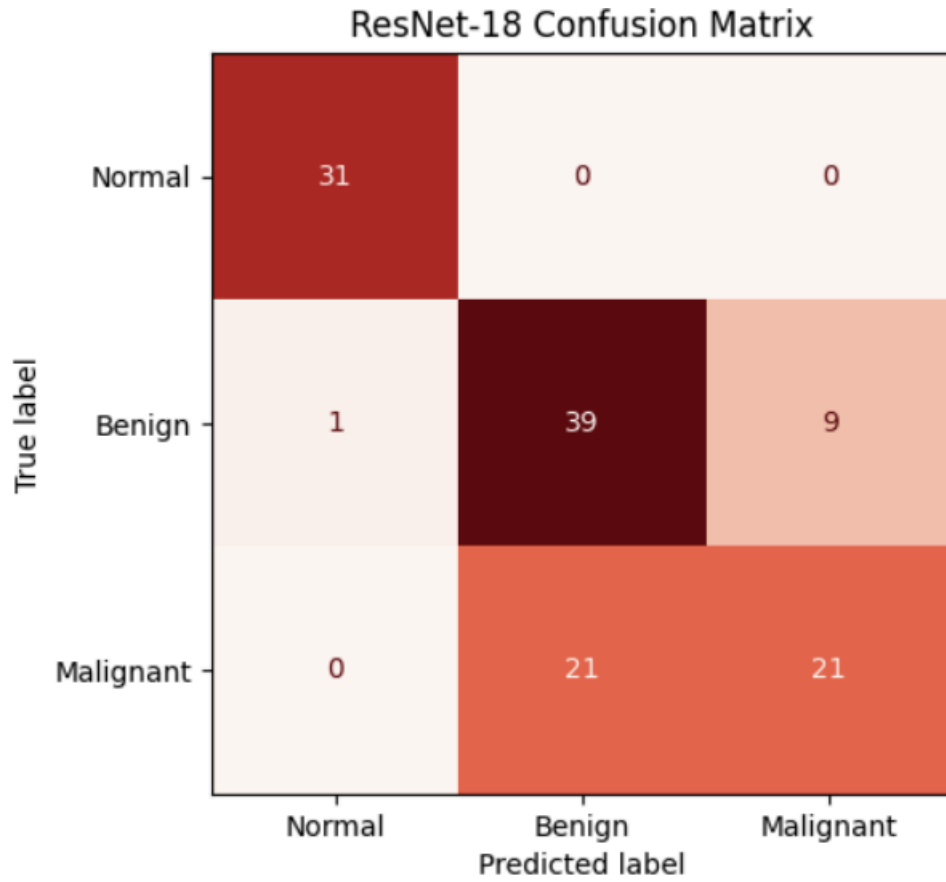


Figure 31: ResNet-18 CNN, 100 Epochs, confusion matrix

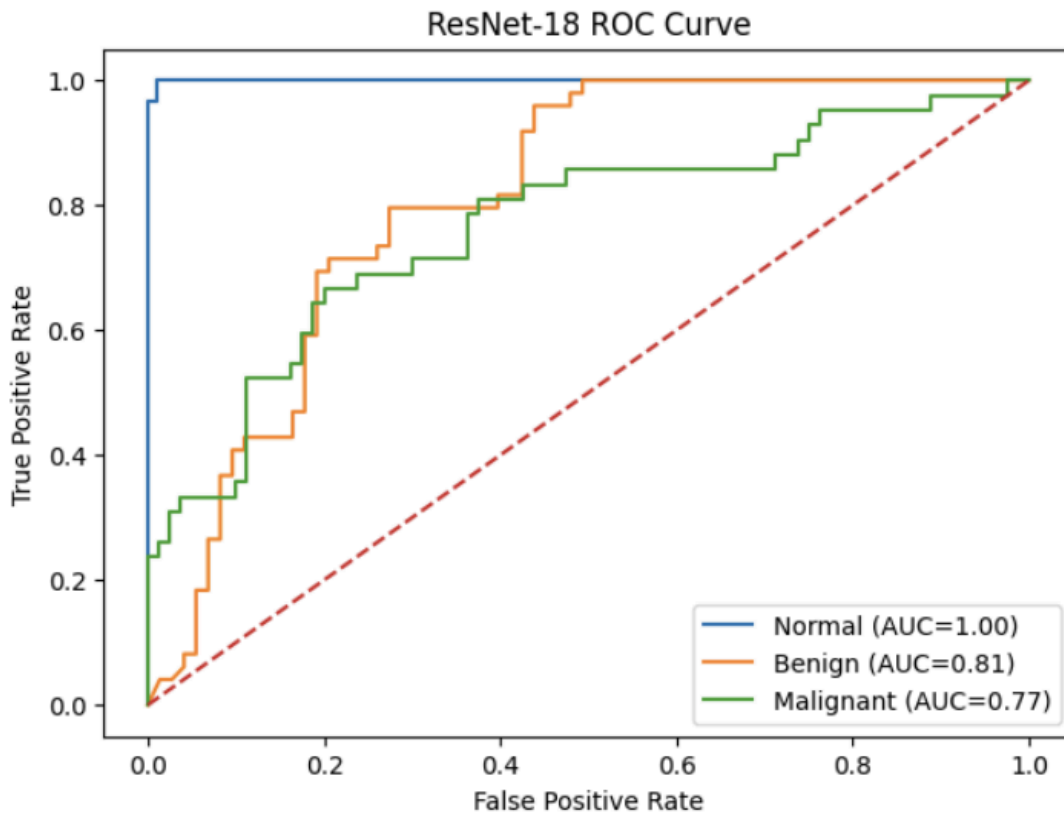


Figure 32: ResNet-18 CNN, 100 Epochs, ROC curve

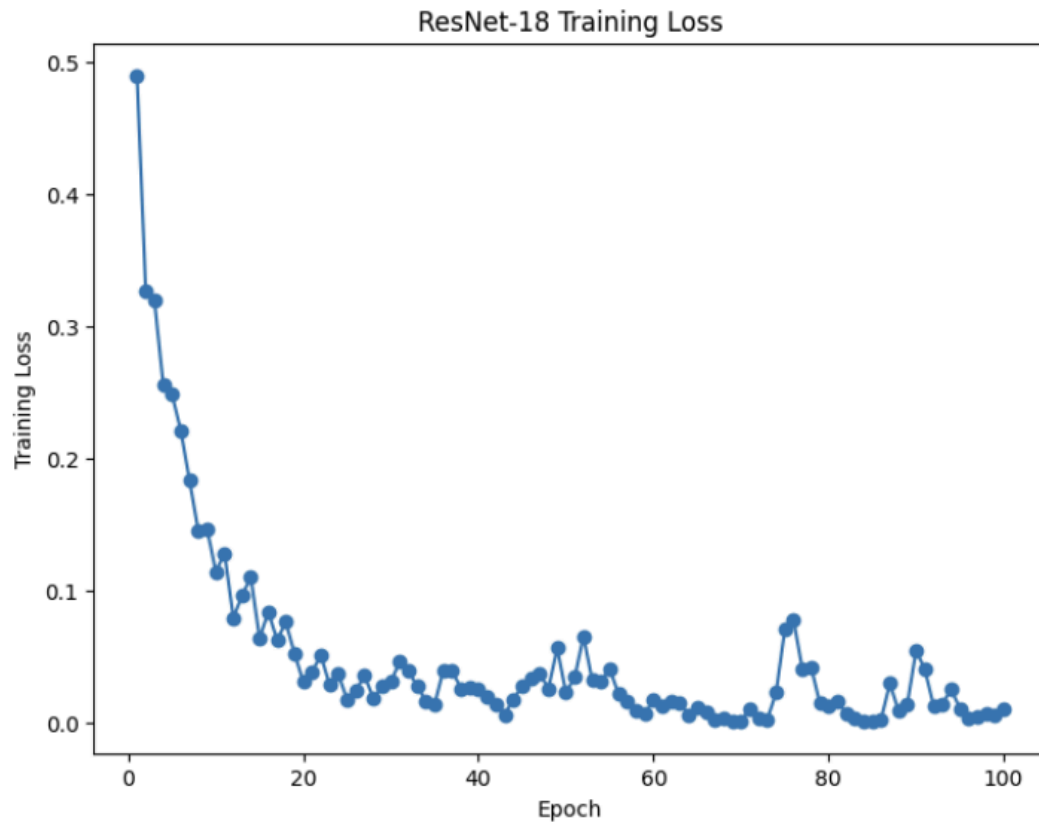


Figure 33: ResNet-18 CNN, Training loss over 100 epochs